BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 4 Note Sheets.

Respiratory · Lymphatic · Alimentary canal · Accessory organs · Abdominal and lower limb vessels · Endocrine

PART A	**The brain dump prompts** All 36 Module 4 competencies, numbers 97 to 132, with two drawing prompts each. Read this part once, then pick.
PART B	**The note sheets** One sheet per competency. The competency and its two prompts on the left, one big drawing box on the right. Thirty six sheets.
PART C	**Check your drawing** For every prompt, what should be on your drawing, and the most common mistake to look for.

How this packet works

Every Module 4 competency is on one sheet with two prompts beside one open box. The number on each sheet is the same number it has in your competency packet. You build one prompt in the box. The other is there if you want more practice, and it is a good place to go the week before the exam.

Before you start

1. Pick two pens you can tell apart. Pass one is the course notes, before any video. Pass two is a second color, after the concept video, into the same box.
2. Stop pass one while part of the box is still empty. That empty part is the point. It is what the video fills in.
3. Read your prompt first, then divide the box to fit it. Some prompts want one labeled drawing. Some want a table. Some want two views side by side. The division is yours to make.
4. Words are allowed inside the drawing, as labels and inside little boxes with arrows between them. A sentence running the width of the box is not. If a box reads like a paragraph, redo it as a picture.

Why two prompts

A and B are never the same question twice. One usually asks for a labeled drawing and the other for a table or a sequence, so a student who thinks in pictures and a student who thinks in categories both get a way in. Neither is the easy option.

About the paper

You do not have to print this. Rule one box per competency onto your own paper, or set the same thing up on a tablet, and copy the prompt letter you chose at the top. A hand ruled sheet counts exactly the same as a printed one.

Lecture and lab

Each competency is tagged Lecture, Lab, or both, the same as your competency packet. Lab tagged competencies are on the lab practical, so practice those on the models and the cadaver too.

PART A

The brain dump prompts.

Thirty six competencies, grouped by topic in the order of your competency packet. Each one gives you two prompts. Build one of them in that competency's box on your note sheet in Part B.

The Respiratory System

97 Upper and lower tracts, conducting and respiratory zones

MODULE 4 · LECTURE · LAB

You should be able to: Distinguish the upper and lower respiratory tracts and the conducting and respiratory zones, and assign the major airway structures to each.

A LABELED AIRWAY OUTLINE

Draw an outline of the head, neck, and chest with the airway from the nose to the lungs. Label the nose, nasal cavity, paranasal sinuses, pharynx, larynx, trachea, bronchi, and lungs. Bracket the upper and lower tracts on the left side and the conducting and respiratory zones on the right. Add a zoomed box of a terminal bronchiole, respiratory bronchiole, alveolar duct, and alveoli.

Then: Circle the spot where the conducting zone ends, and name the airway on each side of that line.

B SORTING GRID

Draw a grid with two columns, conducting zone and respiratory zone, and two rows, upper tract and lower tract. Write each structure in its box: nose, nasal cavity, paranasal sinuses, pharynx, larynx, trachea, bronchi, bronchioles, terminal bronchioles, respiratory bronchioles, alveolar ducts, and alveoli. Add a tiny sketch beside each one.

Then: Write in the box that stays empty why nothing goes there, and list the three things the conducting zone does to air.

98 Nose, nasal cavity, and pharynx

MODULE 4 · LECTURE · LAB

You should be able to: Identify the parts of the nasal cavity, paranasal sinuses, and the three regions of the pharynx.

A LABELED SAGITTAL VIEW

Draw a midsagittal view of the head as it looks on the model. Label the nares, nasal vestibule, the superior, middle, and inferior conchae with the meatus under each, hard palate, soft palate, choanae, and the frontal and sphenoidal sinuses. Bracket the nasopharynx, oropharynx, and laryngopharynx, and label the pharyngeal tonsil, auditory tube opening, palatine tonsil, and lingual tonsil.

Then: Draw a small front view of the face, place the frontal, ethmoid, and maxillary sinuses in their bones, and note which sinus hides behind them.

B FLOW CHAIN

Draw a chain of boxes following air from the nares: nasal vestibule, nasal cavity, choanae, nasopharynx, oropharynx, laryngopharynx. In the nasal cavity box, sketch the nasal septum (septal cartilage, vomer, perpendicular plate of the ethmoid), the three conchae, the meatuses, and the hard and soft palate as the floor. Branch the frontal, ethmoid, sphenoid, and maxillary sinuses into it.

Then: Write under each pharynx box whether it carries air only or air and food, its epithelium, and any tonsils it holds.

99 Larynx

MODULE 4 · LECTURE · LAB

You should be able to: Identify the cartilages and folds of the larynx, including the thyroid and cricoid cartilages, epiglottis, arytenoids, vocal cords, and glottis.

A TWO LABELED VIEWS

Draw two views of the larynx. In the anterior view, label the hyoid bone, epiglottis, thyroid cartilage with the laryngeal prominence, cricothyroid ligament, and cricoid cartilage. In the superior view, looking down into the airway, label the epiglottis, vestibular folds, vocal folds, glottis, and the arytenoid, corniculate, and cuneiform cartilages.

Then: Color the epiglottis to show it is elastic cartilage, and star the only complete ring of cartilage in the airway.

B CARTILAGE AND FOLD TABLE

Draw a table with the thyroid, cricoid, epiglottis, arytenoid, corniculate, and cuneiform cartilages, the vocal folds, the vestibular folds, and the glottis down the side. Across the top: single, paired, or fold, and its shape or where it sits. Use a word or a tiny sketch in each cell.

Then: Sketch the vocal ligament beside the vocal fold row, running from the arytenoid to the thyroid cartilage, and mark which folds make sound.

100 Trachea and bronchial tree

MODULE 4 · LECTURE · LAB

You should be able to: Trace air through the trachea and bronchial tree in order from the trachea and carina down to the terminal bronchioles.

A LABELED BRONCHIAL TREE

Draw the trachea and bronchial tree in anterior view. Label the tracheal cartilages (C rings), carina, right and left main bronchi, lobar bronchi, segmental bronchi, bronchioles, and terminal bronchioles. Draw the right main bronchus wider and more vertical than the left. Number each level in order from the trachea down.

Then: Draw a cross section of the trachea beside the tree showing a C ring, the trachealis muscle, and the esophagus behind it.

B FLOW CHAIN

Draw a chain of boxes from the trachea to the terminal bronchioles: trachea, carina, main bronchi, lobar bronchi, segmental bronchi, bronchioles, terminal bronchioles. Under each, write its other name if it has one (primary, secondary, tertiary) and what it serves (each lung, each lobe, each segment). Add a tiny sketch of its wall.

Then: Draw a bar under the chain showing cartilage shrinking and smooth muscle growing, and mark where the cartilage disappears.

101 Lungs and pleurae

MODULE 4 · LECTURE · LAB

You should be able to: Identify the lobes, fissures, and surfaces of each lung and the parietal and visceral pleurae with the pleural cavity.

A LABELED LUNG VIEWS

Draw the right and left lungs side by side in anterior view. Label the apex, base, superior, middle, and inferior lobes of the right lung, superior and inferior lobes of the left, the horizontal and oblique fissures, the cardiac notch, and the lingula. Mark the hilum on the medial surface of each lung.

Then: Wrap one lung in two lines for the parietal and visceral pleurae, and label the pleural cavity between them.

B COMPARE TABLE

Draw a table with the right lung and the left lung across the top. Down the side: number of lobes, names of the lobes, fissures, special features, and size. Use a word or a tiny sketch in each cell.

Then: Draw a lung inside its two pleural layers below the table and label the parietal pleura, visceral pleura, pleural cavity, apex, base, and hilum.

102 Respiratory zone and alveolar structure

MODULE 4 · LECTURE · LAB

You should be able to: Describe the structure of an alveolus and identify the type I and type II alveolar cells, alveolar macrophages, respiratory membrane, and pulmonary capillaries.

A LABELED ALVEOLAR CLOSE-UP

Draw a respiratory bronchiole leading into an alveolar duct and an alveolar sac, with pulmonary capillaries wrapped around the alveoli. Zoom in on one alveolar wall and label the type I alveolar cells, a type II alveolar cell, an alveolar macrophage on the surface, and a pulmonary capillary.

Then: Draw a tiny inset of the respiratory membrane and label its three layers from the air side to the blood side.

B CELL AND STRUCTURE TABLE

Draw a table with type I alveolar cells, type II alveolar cells, alveolar macrophages, the respiratory membrane, and pulmonary capillaries down the side. Across the top: its shape or what it is made of, where it sits, and its job. Use a word or a tiny sketch in each cell.

Then: Draw one alveolus wrapped in capillaries below the table and place each row of the table on it.

103 Thoracic wall and diaphragm

MODULE 4 · LECTURE · LAB

You should be able to: Identify the diaphragm, the intercostal muscles, and the three structures passing through the diaphragm at T8, T10, and T12.

A LABELED THORAX VIEW

Draw an anterior view of the opened thorax as it looks on the cadaver. Label the ribs, the external and internal intercostals between them, the mediastinum, and the dome-shaped diaphragm between the thoracic and abdominal cavities. Below it, draw the diaphragm from above and place the inferior vena cava at T8, the esophagus at T10, and the aorta at T12.

Then: Draw short lines for the fiber direction of each intercostal layer: hands in pockets for external, praying hands for internal.

B VERTEBRAL LEVEL MAP

Draw a side view of the vertebral column from T6 to T12 with the diaphragm arching in front of it. Draw the inferior vena cava, the esophagus, and the aorta passing through the diaphragm, and line each one up with its level: T8, T10, and T12. Write I-8, E-10, A-12 beside the column.

Then: Add two ribs above with the external and internal intercostals between them, and label the cavity above and below the diaphragm.

104 Respiratory disorders by structure affected

MODULE 4 · LECTURE

You should be able to: Name common respiratory disorders and state which airway, alveolar, or pleural structure each one affects.

A DISORDER MAP

Draw the airway from the trachea down to the bronchioles, one lung inside its pleurae, and a close-up of an alveolar cluster. Place each disorder on the structure it affects, with a tiny sketch of what changes: asthma, COPD, emphysema, pneumonia, pneumothorax, and lung cancer.

Then: Draw an inhaled object in the main bronchus where it most often lands, and write why it lands there.

B DISORDER TABLE

Draw a table with asthma, COPD, emphysema, pneumonia, pneumothorax, and lung cancer down the side. Across the top: the structure affected and what happens to it. Use a word or a tiny sketch in each cell.

Then: Star the disorder that permanently changes the structure of the alveoli, and draw small before and after alveoli beside it.

MODULE 4

The Lymphatic System

105 Lymph pathway

MODULE 4 · LECTURE

You should be able to: Define lymph and trace its one-way route from lymphatic capillaries through collecting vessels, nodes, trunks, and ducts back to the subclavian veins.

A FLOW MAP

Draw a blood capillary bed with tissue cells around it. Show fluid leaking out of the blood capillary into the tissue spaces and a lymphatic capillary picking it up. Then draw a chain of boxes with arrows: lymphatic capillary, collecting vessel, lymph node, trunk, duct, subclavian vein.

Then: Label the interstitial fluid and the lymph, and write beside the lymphatic capillary the moment the fluid becomes lymph.

B DRAINAGE BODY MAP

Draw a body outline. Trace one path from a lymphatic capillary in the left foot: a collecting vessel with valves, an inguinal node, a lumbar trunk, the cisterna chyli, and the thoracic duct emptying where the left subclavian and internal jugular veins meet. Shade the area the right lymphatic duct drains.

Then: Add arrowheads along the whole path to show lymph flows one way, toward the heart, and write a one-line definition of lymph.

106 Vessel and node structure

MODULE 4 · LECTURE · LAB

You should be able to: Describe the structure of lymphatic capillaries, valved collecting vessels, and a lymph node including capsule, cortex, medulla, hilum, and afferent and efferent vessels.

A LABELED NODE

Draw a lymphatic capillary with its blind end, overlapping flap valves, and anchoring filaments, feeding a collecting vessel cut open to show its valves. Follow it into a bean-shaped lymph node. Label the capsule, trabeculae, subcapsular sinus, cortex with follicles, medulla with medullary cords and sinuses, hilum, afferent vessels, and efferent vessels.

Then: Add arrows for the path of lymph through the node, and write beside the hilum why lymph slows down inside the node.

B COMPARE TABLE

Draw a table with lymphatic capillary, collecting vessel, and lymph node down the side. Across the top: what its wall or covering is made of, what keeps lymph moving one way, and where lymph goes next. Use a word or a tiny sketch in each cell.

Then: Draw a lymph node below the table and label the capsule, cortex, medulla, hilum, afferent vessels, and efferent vessels.

107 Primary and secondary organs

MODULE 4 · LECTURE

You should be able to: Sort red bone marrow, thymus, lymph nodes, spleen, tonsils, and MALT into primary and secondary lymphoid organs with location and role.

A SORTING BOX

Draw a box with two columns, primary and secondary. Place red bone marrow, the thymus, lymph nodes, the spleen, the tonsils, and MALT in the correct column. For each one, add a tiny sketch, where it is found, and its role in a few words.

Then: Write one line at the top of each column saying what makes an organ primary or secondary.

B BODY MAP

Draw a body outline and place red bone marrow, the thymus, lymph nodes, the spleen, the tonsils, and MALT where each one is found. Color the two primary organs one color and the four secondary organs another. Beside each, write its role in a few words.

Then: Draw arrows from the red bone marrow and the thymus to the secondary organs to show where mature lymphocytes go.

108 Lymphatic disorders

MODULE 4 · LECTURE

You should be able to: Name edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis and identify the structure each one affects.

A DISORDER MAP

Draw a body outline with a tissue space, a lymphatic vessel in one leg, the nodes of the neck, the spleen, and the tonsils. Place each disorder at the structure it affects, with a tiny sketch of what changes: edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis.

Then: Circle the two disorders that are swelling from fluid, and write how edema differs from lymphedema.

B DISORDER TABLE

Draw a table with edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis down the side. Across the top: the structure affected and what happens there. Use a word or a tiny sketch in each cell.

Then: Star the disorder that makes contact sports dangerous during mononucleosis and say which organ is at risk.

The Alimentary Canal

109 Four-layer gut wall

MODULE 4 · LECTURE · LAB

You should be able to: Name the four layers of the alimentary canal wall from lumen outward and state the dominant tissue of each.

A LABELED CROSS SECTION

Draw a cross section of the gut tube as a set of rings around the lumen. From the lumen outward, label the mucosa (epithelium, lamina propria, muscularis mucosae), the submucosa, the muscularis (inner circular and outer longitudinal layers), and the serosa. Write the main tissue of each layer inside its ring.

Then: Shade the layer that is also called the visceral peritoneum, and write what replaces it on the esophagus.

B COMPARE TABLE

Draw a table with the four wall layers down the side, listed in order from the lumen outward. Across the top: the main tissue of the layer, and the smaller parts found inside it, such as vessels, a nerve plexus, or a thin muscle layer. Use a word or a tiny sketch in each cell.

Then: Write below the table how the outer layer of the esophagus and the muscularis of the stomach break the four-layer pattern.

110 Enteric nerve plexuses

MODULE 4 · LECTURE · LAB

You should be able to: Locate the submucosal and myenteric plexuses in the gut wall and match each to the wall layer it sits within.

A LABELED WALL WEDGE

Draw a wedge of gut wall as it looks on a slide, with the lumen at the bottom. Label the mucosa, submucosa, inner circular muscle, outer longitudinal muscle, and serosa. Draw the submucosal plexus as a dotted network inside the submucosa and the myenteric plexus as a dotted network between the two muscle layers.

Then: Write beside each plexus whether it controls secretion or movement, and name the system the two plexuses make together.

B PLEXUS COMPARE TABLE

Draw a table with the submucosal plexus and the myenteric plexus down the side. Across the top: the wall layer it sits in, the layer just inside it (toward the lumen), the layer just outside it, and what it controls. Use a word or a tiny sketch in each cell.

Then: Write the name of the system the two plexuses form above the table, and circle the plexus that sits within the muscularis.

111 Peritoneum and folds

MODULE 4 · LECTURE · LAB

You should be able to: Identify the parietal and visceral peritoneum, the peritoneal cavity, and the five folds greater omentum, lesser omentum, falciform ligament, mesentery, and mesocolon and state what each fold connects.

A LABELED SAGITTAL VIEW

Draw a side view of the abdomen cut down the middle, with the diaphragm at the top, the anterior wall on one side, and the posterior wall on the other. Draw the liver, stomach, transverse colon, and coils of small intestine. Label the parietal peritoneum, visceral peritoneum, peritoneal cavity, greater omentum, lesser omentum, falciform ligament, mesentery, and mesocolon.

Then: Write beside each of the five folds the two things it connects, or what it hangs from and covers.

B CONNECTION MAP

Draw boxes for the anterior abdominal wall, diaphragm, posterior abdominal wall, liver, stomach, duodenum, jejunum and ileum, transverse colon, sigmoid colon, and intestines. For each of the five folds, draw a line between the boxes it connects and write the fold's name on the line.

Then: Draw two curved lines below the map for the parietal and visceral peritoneum, and label the peritoneal cavity between them.

112 Retroperitoneal organs

MODULE 4 · LECTURE · LAB

You should be able to: Distinguish retroperitoneal organs from intraperitoneal ones and name the kidneys, pancreas, duodenum, and parts of the colon as retroperitoneal.

A LABELED CROSS SECTION

Draw a cross section through the abdomen with the back at the bottom of your box. Draw the parietal peritoneum lining the posterior wall. Place the kidneys, pancreas, duodenum, and a retroperitoneal part of the colon behind it. Then draw a loop of jejunum hanging into the peritoneal cavity on a mesentery, wrapped in visceral peritoneum.

Then: Color every retroperitoneal organ the same color, and write beside the jejunum why it is not retroperitoneal.

B SORTING BOX

Draw a box with two columns: behind the peritoneum, and inside the peritoneal cavity on a fold. Sort these organs: kidneys, pancreas, duodenum, ascending colon, descending colon, stomach, liver, jejunum, ileum, transverse colon, and sigmoid colon. Beside each organ in the second column, write the fold that holds it.

Then: Write across the top of the box, in your own words, what makes an organ retroperitoneal.

113 Mouth, pharynx, esophagus

MODULE 4 · LECTURE · LAB

You should be able to: Identify the oral cavity boundaries, hard and soft palate, uvula, fauces, tonsils, the three pharyngeal regions, and the esophagus with its hiatus.

A LABELED SAGITTAL VIEW

Draw a side view of the head and neck cut down the middle. Label the oral cavity, hard palate, soft palate, uvula, fauces, palatine tonsil, lingual tonsil, nasopharynx, oropharynx, laryngopharynx, and esophagus. Continue the esophagus down to a line for the diaphragm and label the esophageal hiatus.

Then: Write the boundaries of the oral cavity beside the mouth, and draw the uvula lifted to close off the nasopharynx.

B FRONT VIEW AND CHAIN

Draw the open mouth from the front. Label the lips, cheeks, hard palate, soft palate, uvula, fauces, the palatine tonsils between the palatal arches, and the tongue with the lingual tonsil at its base. Beside it, draw boxes for the path of food from the fauces through the oropharynx, laryngopharynx, esophagus, and esophageal hiatus to the stomach, with a side box for the nasopharynx.

Then: Write beside your chain the two bones that form the hard palate and the length of the esophagus.

114 Major canal sphincters

MODULE 4 · LECTURE · LAB

You should be able to: Locate the upper and lower esophageal, pyloric, ileocecal, and internal and external anal sphincters and state which regions each one separates.

A LABELED CANAL OUTLINE

Draw a simple outline of the alimentary canal from the pharynx to the anus. Draw each sphincter as a ring and label it: upper esophageal, lower esophageal (cardiac), pyloric, ileocecal, internal anal, and external anal. Label the region on each side of every ring.

Then: Write the muscle type beside the pyloric, ileocecal, internal anal, and external anal sphincters, and color the one you control by choice.

B SPHINCTER TABLE

Draw a table with the six sphincters down the side, in the order food meets them. Across the top: the region just before it, the region just after it, and any second name it has. Use a word or a tiny sketch in each cell.

Then: Star the two sphincters that guard the anus, and write which one you control by choice.

115 Stomach regions and features

MODULE 4 · LECTURE · LAB

You should be able to: Identify the cardia, fundus, body, pyloric part, curvatures, rugae, and the three-layered muscularis of the stomach.

A LABELED ANTERIOR VIEW

Draw the J-shaped stomach below the diaphragm with the esophagus entering and the duodenum leaving. Label the cardia, fundus, body, pyloric part (pyloric antrum, pyloric canal, pylorus), pyloric sphincter, and the lesser and greater curvatures. Cut a window in the front wall to show the rugae, and peel back the muscle layers at one edge.

Then: Label the three muscle layers in order from inside to outside, and write which layer the rest of the tube does not have.

B REGION TABLE

Draw a table with the cardia, fundus, body, pyloric part, lesser curvature, greater curvature, and rugae down the side. Across the top: where it is on the stomach and what it looks like. Use a word or a tiny sketch in each cell.

Then: Draw a strip of stomach wall below the table, and label its inner oblique, middle circular, and outer longitudinal muscle layers.

116 Small intestine and surface area

MODULE 4 · LECTURE · LAB

You should be able to: Identify the duodenum, jejunum, and ileum and name the circular folds, villi, lacteal, and microvilli as features that expand absorptive surface area.

A LABELED MODEL VIEW

Draw the small intestine as it sits in the abdomen, from the pyloric sphincter to the ileocecal sphincter. Label the duodenum, jejunum, and ileum with the length of each, and the mesentery on the jejunum and ileum. Off the jejunum, draw a zoom bubble showing a circular fold, one villus with its lacteal, and microvilli on one absorptive cell.

Then: Shade the region that is mostly retroperitoneal, and number the three surface features from largest to smallest.

B ZOOM SEQUENCE

Draw a straight strip divided into the duodenum, jejunum, and ileum, with the length of each written on it. From one spot, draw three zooms, each smaller than the last: a circular fold of mucosa and submucosa, one villus with its lacteal, and one absorptive cell with microvilli forming the brush border.

Then: Write beside the lacteal what it absorbs, and beside the ileum the sphincter where it joins the large intestine.

117 Large intestine features

MODULE 4 · LECTURE · LAB

You should be able to: Identify the cecum, appendix, four parts of the colon, rectum, and anal canal along with the teniae coli, haustra, and omental appendices.

A LABELED ANTERIOR VIEW

Draw the large intestine framing the abdomen, with the ileum entering at the ileocecal valve. Label the cecum, vermiform appendix, ascending colon, transverse colon, descending colon, sigmoid colon, rectum, anal canal, and the internal and external anal sphincters. On the transverse colon, draw and label the teniae coli, haustra, and omental appendices.

Then: Write beside the rectum and the anal canal where each one sits, and color the two colon parts that hang on a mesocolon.

B REGION TABLE

Draw a table with the cecum, the four parts of the colon, the rectum, and the anal canal down the side, in the order feces move through them. Across the top: what comes just before it, what comes just after it, and one thing that helps you find it on the model. Use a word or a tiny sketch in each cell.

Then: Draw a cecum with its appendix below the table, and a short piece of colon with three teniae coli, haustra, and omental appendices.

118 Trace the pathway of food

MODULE 4 · LECTURE · LAB

You should be able to: Trace a bite of food through every named segment of the alimentary canal in order from mouth to anus.

A FLOW CHAIN

Draw a chain of boxes with arrows that carries a bite of food from the mouth to the anus. Put every named segment in its own box, including the two parts of the pharynx that food passes, the three parts of the small intestine, and each part of the large intestine. Draw a ring on the arrow wherever a sphincter sits.

Then: Write one word under each box for what happens to the food there, using the pathway in your notes.

B BODY OUTLINE MAP

Draw an outline of the head and trunk with the diaphragm. Draw the whole canal in place, from the mouth through the pharynx, esophagus, stomach, small intestine, and large intestine to the anus. Number every named segment in the order food reaches it, and label each number.

Then: Circle the region where most absorption happens, and box the region where water is absorbed and feces form.

119 Clinical anatomy of canal disorders

MODULE 4 · LECTURE

You should be able to: Match GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, IBD, hemorrhoids, and colorectal cancer to the canal structure each one affects.

A DISORDER MAP

Draw an outline of the canal from the esophagus to the anus, with the diaphragm and its esophageal hiatus. Write each disorder on the structure it affects: GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, inflammatory bowel disease, hemorrhoids, and colorectal cancer. Draw a small sketch of the problem beside each label, such as a pouch or a swollen vein.

Then: Write beside the lower esophageal sphincter and the appendix where the patient feels the pain from each problem.

B COMPARE TABLE

Draw a table with the eight disorders down the side: GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, inflammatory bowel disease, hemorrhoids, and colorectal cancer. Across the top: the structure it affects and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Draw the four wall layers beside the table, and show how a peptic ulcer erodes from the mucosa until it perforates into the peritoneal cavity.

MODULE 4

The Accessory Digestive Organs

120 Teeth and tongue

MODULE 4 · LECTURE · LAB

You should be able to: Identify the teeth and tongue with the lingual frenulum, papillae, intrinsic and extrinsic muscles, and lingual glands.

A LABELED MOUTH VIEW

Draw the open mouth from the front with the upper and lower teeth and the tongue lifted toward the roof of the mouth. Label the teeth, the tongue, and the lingual frenulum under it. Beside it, draw a side view of the tongue and label the papillae on its upper surface, the intrinsic muscles, the extrinsic muscles, and the lingual glands.

Then: Draw arrows for the movements the extrinsic muscles make, and write what the intrinsic muscles change about the tongue.

B STRUCTURE TABLE

Draw a table with the teeth, tongue, lingual frenulum, papillae, intrinsic tongue muscles, extrinsic tongue muscles, and lingual glands down the side. Across the top: where it is and what it does. Use a word or a tiny sketch in each cell.

Then: Circle the two muscle groups, and write beside them which group starts outside the tongue and which sits within it.

121 Salivary glands

MODULE 4 · LECTURE · LAB

You should be able to: Locate the parotid, submandibular, and sublingual glands and state where the duct of each opens.

A LABELED FACE VIEW

Draw a side view of the face with the ear, the mandible, the masseter muscle, and the inside of the mouth. Draw and label the parotid, submandibular, and sublingual glands in place. Draw each gland's duct running to the spot where it opens inside the mouth, and label that spot.

Then: Color each gland and its duct opening the same color, and write the muscle the parotid gland lies over.

B LOCATION TABLE

Draw a table with the parotid, submandibular, and sublingual glands down the side. Across the top: where the gland sits, the landmark next to it, and where its duct opens in the mouth. Use a word or a tiny sketch in each cell.

Then: Sketch the floor of the mouth below the table with the tongue lifted, and mark where the submandibular and sublingual ducts open.

122 Liver gross and lobule anatomy

MODULE 4 · LECTURE · LAB

You should be able to: Identify the four liver lobes, falciform ligament, and ligamentum teres, and name the hepatocytes, sinusoids, canaliculi, and portal triad of the lobule.

A LABELED VIEWS AND ZOOM

Draw the liver from the front and label the right lobe, left lobe, falciform ligament, and ligamentum teres. Draw a small view of its posterior and inferior surface with the quadrate and caudate lobes. Add a zoom bubble of one lobule with plates of hepatocytes, sinusoids between the plates, bile canaliculi, and a portal triad at one corner.

Then: Label the three parts of the portal triad, and write the fetal vessel the ligamentum teres is a remnant of.

B GROSS TO MICRO TABLE

Draw a table with the right, left, quadrate, and caudate lobes, the falciform ligament, ligamentum teres, hepatocytes, sinusoids, bile canaliculi, and portal triad down the side. Across the top: seen by eye or on a slide, where it is, and what it is or does. Use a word or a tiny sketch in each cell.

Then: Draw one portal triad below the table, and add arrows for blood coming in and bile going out.

123 Gallbladder and bile pathway

MODULE 4 · LECTURE · LAB

You should be able to: Identify the gallbladder regions and trace bile from the canaliculi through the hepatic, cystic, and common bile ducts to the hepatopancreatic ampulla and duodenum.

A LABELED BILIARY TREE

Draw the underside of the liver with the gallbladder, the duodenum below, and the pancreas beside it. Add a small bubble of bile canaliculi between hepatocytes. Label the fundus, body, and neck of the gallbladder, the right and left hepatic ducts, common hepatic duct, cystic duct, common bile duct, pancreatic duct, hepatopancreatic ampulla, sphincter of Oddi, and major duodenal papilla.

Then: Draw arrows that carry bile from the canaliculi to the duodenum, with a side loop through the cystic duct into the gallbladder.

B FLOW CHAIN

Draw a chain of boxes with arrows that carries a drop of bile from the hepatocyte that made it to the duodenum. Give every duct and opening its own box, starting at the bile canaliculi. Branch a side arrow off the chain to the gallbladder, and sketch the gallbladder with its fundus, body, and neck.

Then: Write on the chain where the pancreatic duct joins and where the sphincter of Oddi controls flow.

124 Pancreas structure

MODULE 4 · LECTURE · LAB

You should be able to: Identify the head, body, and tail of the pancreas with its main and accessory ducts, major duodenal papilla, sphincter of Oddi, acini, and islets.

A LABELED VIEW AND ZOOM

Draw the pancreas behind the stomach with its head in the curve of the duodenum. Label the head, body, and tail, the main pancreatic duct, the accessory duct, the common bile duct joining at the hepatopancreatic ampulla, the sphincter of Oddi, and the major duodenal papilla. Add a zoom bubble with acini and one pancreatic islet.

Then: Write the second name of each duct, and label which cluster is exocrine and which is endocrine, with its percent of the gland.

B COMPARE TABLE

Draw a table with the head, body, tail, main pancreatic duct, accessory duct, major duodenal papilla, sphincter of Oddi, acini, and pancreatic islets down the side. Across the top: where it is, its other name if it has one, and what it does or carries. Use a word or a tiny sketch in each cell.

Then: Sketch the two ducts entering the duodenum below the table, and circle the one that joins the common bile duct.

125 Clinical anatomy of accessory disorders

MODULE 4 · LECTURE

You should be able to: Match gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps to the accessory organ each one affects and explain how a stone at the ampulla can block bile and pancreatic flow at once.

A DISORDER MAP

Draw the liver, gallbladder, bile ducts, pancreas, and duodenum together, and a small side view of the face with the parotid gland. Write each disorder on the organ it affects: gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps. Draw a gallstone stuck at the hepatopancreatic ampulla.

Then: Draw arrows showing bile and pancreatic juice both backed up behind the stone, and write which disorder this can trigger.

B COMPARE TABLE

Draw a table with gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps down the side. Across the top: the organ it affects and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Draw the bile duct and pancreatic duct meeting at the ampulla below the table, place a stone there, and write which two flows stop.

MODULE 4

Lab: Vessels and Nerves of the Abdomen and Lower Limb

126 Abdominal aorta and portal system

MODULE 4 · LAB

You should be able to: Identify the major branches of the abdominal aorta and the vessels of the hepatic portal system on the cadaver.

A LABELED ANTERIOR VIEW

Draw the abdominal aorta and inferior vena cava. Label the celiac trunk with its left gastric, splenic and common hepatic arteries, then the superior mesenteric, renal, gonadal and inferior mesenteric arteries, and the right and left common iliacs. Above, label the liver, hepatic veins, hepatic portal vein, the splenic and superior mesenteric veins that form it, and the inferior mesenteric vein joining the splenic vein.

Then: Number the three unpaired aortic branches 1 to 3 from top to bottom, and write beside the liver what makes this a portal system.

B BRANCH AND DRAINAGE TABLE

Draw a table with these vessels down the side: celiac trunk, left gastric, splenic, common hepatic, superior mesenteric, renal, gonadal and inferior mesenteric arteries, then the hepatic portal, splenic, superior mesenteric, inferior mesenteric and hepatic veins. Across the top: paired or unpaired, and the vessel it branches off or drains into. Use a word in each cell.

Then: Below the table, write the chain of vessels a glucose molecule follows from an intestinal capillary to the right atrium, and circle the two capillary beds.

127 Lower limb vessels and nerves

MODULE 4 · LAB

You should be able to: Identify the femoral and lower-limb blood vessels and the major nerves of the abdomen and lower limb on the cadaver.

A FRONT AND BACK VIEWS

Outline the right lower limb twice, from the front and from the back. On the front view, label the femoral artery, femoral vein, femoral nerve and great saphenous vein in the thigh, and carry the great saphenous vein down the leg. On the back view, label the sciatic nerve, its tibial and common fibular branches, and the posterior tibial artery in the leg.

Then: Write deep or superficial beside each vein, and write lumbar plexus or sacral plexus beside each nerve.

B STRUCTURE-TO-REGION TABLE

Draw a table with the femoral artery, femoral vein, great saphenous vein, posterior tibial artery, femoral nerve, sciatic nerve, tibial nerve and common fibular nerve down the side. Across the top: artery, vein or nerve; found in the thigh, the leg, or both; and deep or superficial (veins only). Use a word or a tiny sketch in each cell.

Then: Below the table, draw two boxes for the lumbar and sacral plexuses and branch each nerve from the plexus or nerve it comes from.

MODULE 4

The Endocrine System

128 Major endocrine glands and locations

MODULE 4 · LECTURE

You should be able to: Name the major endocrine glands and locate each one in the body from the hypothalamus to the gonads.

A LABELED BODY MAP

Draw a body outline from the head to the pelvis. Place and label the hypothalamus, pituitary gland, pineal gland, thyroid gland, parathyroid glands, thymus, adrenal glands, pancreatic islets, ovaries and testes. Beside each one, write its landmark from your notes, such as the third ventricle, the sella turcica, the larynx, the sternum, the kidney, the stomach, the uterus or the scrotum.

Then: Star the largest purely endocrine gland, and write beside the thymus how its size changes after puberty.

B LOCATION TABLE

Draw a table with the ten major endocrine glands down the side, from the hypothalamus to the testes. Across the top: the body region (head, neck, chest, abdomen, pelvis or scrotum), and the bone, organ or space it sits in or against. Use a word or a tiny sketch in each cell.

Then: Circle every gland that comes in a pair or more, and write beside the parathyroid glands how many there usually are.

129 Pituitary gross anatomy and lobes

MODULE 4 · LECTURE · LAB

You should be able to: Describe the pituitary in the sella turcica, its infundibulum link to the hypothalamus, and compare the anterior and posterior lobes by tissue type.

A LABELED SAGITTAL VIEW

Draw a midsagittal view of the base of the brain sitting over the sphenoid bone. Label the hypothalamus, infundibulum, sella turcica, and the pituitary gland (hypophysis) with its anterior lobe (adenohypophysis) and posterior lobe (neurohypophysis). Draw the nerve fibers running down the infundibulum to the posterior lobe and the hypophyseal portal vessels reaching the anterior lobe.

Then: Write the tissue type inside each lobe, then add an arrow showing which way a pituitary tumor grows and label what it presses on.

B TWO-LOBE COMPARE TABLE

Draw a table with the anterior lobe and the posterior lobe down the side. Across the top: its other name, the tissue it is made of, and how it links to the hypothalamus. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch the gland in the sella turcica hanging from the infundibulum, and write one sentence on how the tissue type predicts the link.

130 Thyroid and parathyroid anatomy

MODULE 4 · LECTURE · LAB

You should be able to: Identify the thyroid lobes, isthmus, follicles, colloid, follicular and parafollicular cells, and the parathyroid chief and oxyphil cells.

A LABELED VIEWS WITH INSETS

Draw the thyroid wrapped on the trachea just below the larynx, seen from the front, and label the right lobe, left lobe, isthmus, trachea and larynx. Beside it, draw a back view with the usual four parathyroid glands. Add one magnified circle of thyroid follicles labeling colloid, follicular cells and parafollicular (C) cells, and a second circle of parathyroid tissue labeling chief cells.

Then: Write beside the back view why the parathyroid glands are at risk during thyroid surgery.

B ZOOM-IN SEQUENCE

Draw three boxes in a row, each zooming in on the one before: the neck with the thyroid on the trachea, the gland with its lobes and isthmus, then a group of follicles. In the last box, label colloid, follicular cells in the wall, and parafollicular (C) cells between follicles. Add a side box for a parathyroid gland and label its chief cells.

Then: Write under each box whether you would identify it on the model, the cadaver, or the slide.

131 Adrenal gland zones

MODULE 4 · LECTURE · LAB

You should be able to: Identify the adrenal cortex zones in order and the adrenal medulla with its chromaffin cells.

A LABELED CUTAWAY

Draw a kidney with the adrenal gland capping its superior pole, then cut a wedge out of the adrenal to show its inside. Label the capsule, the adrenal cortex with its three zones in order from the surface inward (zona glomerulosa, zona fasciculata, zona reticularis), and the adrenal medulla at the core. Draw each zone at its relative thickness.

Then: Write the tissue type beside the cortex and beside the medulla, and sketch the cell pattern of the fasciculata and the reticularis.

B LAYER TABLE

Draw a table with the zona glomerulosa, zona fasciculata, zona reticularis and adrenal medulla down the side, in order from the surface inward. Across the top: its position (outer, middle, inner or core), its thickness, and its tissue type. Use a word or a tiny sketch in each cell.

Then: Below the table, write why imaging ordered for a kidney often shows the adrenal gland as well.

132 Pancreas, gonads, and organs with endocrine tissue

MODULE 4 · LECTURE

You should be able to: Locate the pancreatic islets and gonads and name other organs such as the heart, kidneys, and skin that contain endocrine tissue.

A LABELED BODY MAP

Draw a body outline. Draw the pancreas behind the stomach with a magnified circle of pancreatic islets among the acini, the ovaries beside the uterus, and the testes in the scrotum. Then star the heart, kidneys, skin, stomach and small intestine, liver, adipose tissue and placenta, and label the endocrine tissue each one contains, such as cells in the wall of the atria.

Then: Write endocrine on the islets and exocrine on the acini, and write beside the placenta what makes it a temporary organ.

B ORGAN SORTING TABLE

Draw a table with the pancreas, ovaries, testes, heart, kidneys, skin, stomach and small intestine, liver, adipose tissue and placenta down the side. Across the top: where it sits, and the endocrine tissue it contains. Use a word or a tiny sketch in each cell.

Then: Circle the three organs your notes call mixed organs, and write beside them what makes an organ mixed.

PART B

The note sheets.

One sheet per competency, 36 sheets. The competency and both prompts sit on the left so nothing else has to be printed beside it. The box on the right is the whole assignment.

The box is **drawn**, not written. Pick one prompt and divide the box however it needs. Words go inside the drawing, in labels and little boxes with arrows between them. Sentences running across the page do not count.

97
COMP

MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Upper and lower tracts, conducting and respiratory zones

You should be able to: Distinguish the upper and lower respiratory tracts and the conducting and respiratory zones, and assign the major airway structures to each.

PICK ONE AND DO THAT ONE

☐ **A** LABELED AIRWAY OUTLINE

Draw an outline of the head, neck, and chest with the airway from the nose to the lungs. Label the nose, nasal cavity, paranasal sinuses, pharynx, larynx, trachea, bronchi, and lungs. Bracket the upper and lower tracts on the left side and the conducting and respiratory zones on the right. Add a zoomed box of a terminal bronchiole, respiratory bronchiole, alveolar duct, and alveoli.

Then: Circle the spot where the conducting zone ends, and name the airway on each side of that line.

☐ **B** SORTING GRID

Draw a grid with two columns, conducting zone and respiratory zone, and two rows, upper tract and lower tract. Write each structure in its box: nose, nasal cavity, paranasal sinuses, pharynx, larynx, trachea, bronchi, bronchioles, terminal bronchioles, respiratory bronchioles, alveolar ducts, and alveoli. Add a tiny sketch beside each one.

Then: Write in the box that stays empty why nothing goes there, and list the three things the conducting zone does to air.

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PASS 2 COLOR _____

DRAW IT

split it up however the prompt needs

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98

COMP

MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Nose, nasal cavity, and pharynx

You should be able to: Identify the parts of the nasal cavity, paranasal sinuses, and the three regions of the pharynx.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SAGITTAL VIEW

Draw a midsagittal view of the head as it looks on the model. Label the nares, nasal vestibule, the superior, middle, and inferior conchae with the meatus under each, hard palate, soft palate, choanae, and the frontal and sphenoidal sinuses. Bracket the nasopharynx, oropharynx, and laryngopharynx, and label the pharyngeal tonsil, auditory tube opening, palatine tonsil, and lingual tonsil.

Then: Draw a small front view of the face, place the frontal, ethmoid, and maxillary sinuses in their bones, and note which sinus hides behind them.

☐ **B** FLOW CHAIN

Draw a chain of boxes following air from the nares: nasal vestibule, nasal cavity, choanae, nasopharynx, oropharynx, laryngopharynx. In the nasal cavity box, sketch the nasal septum (septal cartilage, vomer, perpendicular plate of the ethmoid), the three conchae, the meatuses, and the hard and soft palate as the floor. Branch the frontal, ethmoid, sphenoid, and maxillary sinuses into it.

Then: Write under each pharynx box whether it carries air only or air and food, its epithelium, and any tonsils it holds.

DRAW IT

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COMP

MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Larynx

You should be able to: Identify the cartilages and folds of the larynx, including the thyroid and cricoid cartilages, epiglottis, arytenoids, vocal cords, and glottis.

PICK ONE AND DO THAT ONE

☐ **A** TWO LABELED VIEWS

Draw two views of the larynx. In the anterior view, label the hyoid bone, epiglottis, thyroid cartilage with the laryngeal prominence, cricothyroid ligament, and cricoid cartilage. In the superior view, looking down into the airway, label the epiglottis, vestibular folds, vocal folds, glottis, and the arytenoid, corniculate, and cuneiform cartilages.

Then: Color the epiglottis to show it is elastic cartilage, and star the only complete ring of cartilage in the airway.

☐ **B** CARTILAGE AND FOLD TABLE

Draw a table with the thyroid, cricoid, epiglottis, arytenoid, corniculate, and cuneiform cartilages, the vocal folds, the vestibular folds, and the glottis down the side. Across the top: single, paired, or fold, and its shape or where it sits. Use a word or a tiny sketch in each cell.

Then: Sketch the vocal ligament beside the vocal fold row, running from the arytenoid to the thyroid cartilage, and mark which folds make sound.

DRAW IT

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COMP

MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Trachea and bronchial tree

You should be able to: Trace air through the trachea and bronchial tree in order from the trachea and carina down to the terminal bronchioles.

PICK ONE AND DO THAT ONE

☐ **A** LABELED BRONCHIAL TREE

Draw the trachea and bronchial tree in anterior view. Label the tracheal cartilages (C rings), carina, right and left main bronchi, lobar bronchi, segmental bronchi, bronchioles, and terminal bronchioles. Draw the right main bronchus wider and more vertical than the left. Number each level in order from the trachea down.

Then: Draw a cross section of the trachea beside the tree showing a C ring, the trachealis muscle, and the esophagus behind it.

☐ **B** FLOW CHAIN

Draw a chain of boxes from the trachea to the terminal bronchioles: trachea, carina, main bronchi, lobar bronchi, segmental bronchi, bronchioles, terminal bronchioles. Under each, write its other name if it has one (primary, secondary, tertiary) and what it serves (each lung, each lobe, each segment). Add a tiny sketch of its wall.

Then: Draw a bar under the chain showing cartilage shrinking and smooth muscle growing, and mark where the cartilage disappears.

DRAW IT

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101
COMP

MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Lungs and pleurae

You should be able to: Identify the lobes, fissures, and surfaces of each lung and the parietal and visceral pleurae with the pleural cavity.

PICK ONE AND DO THAT ONE

☐ **A** LABELED LUNG VIEWS

Draw the right and left lungs side by side in anterior view. Label the apex, base, superior, middle, and inferior lobes of the right lung, superior and inferior lobes of the left, the horizontal and oblique fissures, the cardiac notch, and the lingula. Mark the hilum on the medial surface of each lung.

Then: Wrap one lung in two lines for the parietal and visceral pleurae, and label the pleural cavity between them.

☐ **B** COMPARE TABLE

Draw a table with the right lung and the left lung across the top. Down the side: number of lobes, names of the lobes, fissures, special features, and size. Use a word or a tiny sketch in each cell.

Then: Draw a lung inside its two pleural layers below the table and label the parietal pleura, visceral pleura, pleural cavity, apex, base, and hilum.

DRAW IT

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102
COMP

MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Respiratory zone and alveolar structure

You should be able to: Describe the structure of an alveolus and identify the type I and type II alveolar cells, alveolar macrophages, respiratory membrane, and pulmonary capillaries.

PICK ONE AND DO THAT ONE

☐ **A** LABELED ALVEOLAR CLOSE-UP

Draw a respiratory bronchiole leading into an alveolar duct and an alveolar sac, with pulmonary capillaries wrapped around the alveoli. Zoom in on one alveolar wall and label the type I alveolar cells, a type II alveolar cell, an alveolar macrophage on the surface, and a pulmonary capillary.

Then: Draw a tiny inset of the respiratory membrane and label its three layers from the air side to the blood side.

☐ **B** CELL AND STRUCTURE TABLE

Draw a table with type I alveolar cells, type II alveolar cells, alveolar macrophages, the respiratory membrane, and pulmonary capillaries down the side. Across the top: its shape or what it is made of, where it sits, and its job. Use a word or a tiny sketch in each cell.

Then: Draw one alveolus wrapped in capillaries below the table and place each row of the table on it.

DRAW IT

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103
COMP

MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE AND LAB

Thoracic wall and diaphragm

You should be able to: Identify the diaphragm, the intercostal muscles, and the three structures passing through the diaphragm at T8, T10, and T12.

PICK ONE AND DO THAT ONE

☐ **A** LABELED THORAX VIEW

Draw an anterior view of the opened thorax as it looks on the cadaver. Label the ribs, the external and internal intercostals between them, the mediastinum, and the dome-shaped diaphragm between the thoracic and abdominal cavities. Below it, draw the diaphragm from above and place the inferior vena cava at T8, the esophagus at T10, and the aorta at T12.

Then: Draw short lines for the fiber direction of each intercostal layer: hands in pockets for external, praying hands for internal.

☐ **B** VERTEBRAL LEVEL MAP

Draw a side view of the vertebral column from T6 to T12 with the diaphragm arching in front of it. Draw the inferior vena cava, the esophagus, and the aorta passing through the diaphragm, and line each one up with its level: T8, T10, and T12. Write I-8, E-10, A-12 beside the column.

Then: Add two ribs above with the external and internal intercostals between them, and label the cavity above and below the diaphragm.

DRAW IT

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104
COMP

MODULE 4 · THE RESPIRATORY SYSTEM · LECTURE

Respiratory disorders by structure affected

You should be able to: Name common respiratory disorders and state which airway, alveolar, or pleural structure each one affects.

PICK ONE AND DO THAT ONE

☐ **A** DISORDER MAP

Draw the airway from the trachea down to the bronchioles, one lung inside its pleurae, and a close-up of an alveolar cluster. Place each disorder on the structure it affects, with a tiny sketch of what changes: asthma, COPD, emphysema, pneumonia, pneumothorax, and lung cancer.

Then: Draw an inhaled object in the main bronchus where it most often lands, and write why it lands there.

☐ **B** DISORDER TABLE

Draw a table with asthma, COPD, emphysema, pneumonia, pneumothorax, and lung cancer down the side. Across the top: the structure affected and what happens to it. Use a word or a tiny sketch in each cell.

Then: Star the disorder that permanently changes the structure of the alveoli, and draw small before and after alveoli beside it.

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105
COMP

MODULE 4 · THE LYMPHATIC SYSTEM · LECTURE

Lymph pathway

You should be able to: Define lymph and trace its one-way route from lymphatic capillaries through collecting vessels, nodes, trunks, and ducts back to the subclavian veins.

PICK ONE AND DO THAT ONE

☐ **A** FLOW MAP

Draw a blood capillary bed with tissue cells around it. Show fluid leaking out of the blood capillary into the tissue spaces and a lymphatic capillary picking it up. Then draw a chain of boxes with arrows: lymphatic capillary, collecting vessel, lymph node, trunk, duct, subclavian vein.

Then: Label the interstitial fluid and the lymph, and write beside the lymphatic capillary the moment the fluid becomes lymph.

☐ **B** DRAINAGE BODY MAP

Draw a body outline. Trace one path from a lymphatic capillary in the left foot: a collecting vessel with valves, an inguinal node, a lumbar trunk, the cisterna chyli, and the thoracic duct emptying where the left subclavian and internal jugular veins meet. Shade the area the right lymphatic duct drains.

Then: Add arrowheads along the whole path to show lymph flows one way, toward the heart, and write a one-line definition of lymph.

DRAW IT

split it up however the prompt needs

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106
COMP

MODULE 4 · THE LYMPHATIC SYSTEM · LECTURE AND LAB

Vessel and node structure

You should be able to: Describe the structure of lymphatic capillaries, valved collecting vessels, and a lymph node including capsule, cortex, medulla, hilum, and afferent and efferent vessels.

PICK ONE AND DO THAT ONE

☐ **A** LABELED NODE

Draw a lymphatic capillary with its blind end, overlapping flap valves, and anchoring filaments, feeding a collecting vessel cut open to show its valves. Follow it into a bean-shaped lymph node. Label the capsule, trabeculae, subcapsular sinus, cortex with follicles, medulla with medullary cords and sinuses, hilum, afferent vessels, and efferent vessels.

Then: Add arrows for the path of lymph through the node, and write beside the hilum why lymph slows down inside the node.

☐ **B** COMPARE TABLE

Draw a table with lymphatic capillary, collecting vessel, and lymph node down the side. Across the top: what its wall or covering is made of, what keeps lymph moving one way, and where lymph goes next. Use a word or a tiny sketch in each cell.

Then: Draw a lymph node below the table and label the capsule, cortex, medulla, hilum, afferent vessels, and efferent vessels.

DRAW IT

split it up however the prompt needs

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COMP

MODULE 4 · THE LYMPHATIC SYSTEM · LECTURE

Primary and secondary organs

You should be able to: Sort red bone marrow, thymus, lymph nodes, spleen, tonsils, and MALT into primary and secondary lymphoid organs with location and role.

PICK ONE AND DO THAT ONE

☐ **A** SORTING BOX

Draw a box with two columns, primary and secondary. Place red bone marrow, the thymus, lymph nodes, the spleen, the tonsils, and MALT in the correct column. For each one, add a tiny sketch, where it is found, and its role in a few words.

Then: Write one line at the top of each column saying what makes an organ primary or secondary.

☐ **B** BODY MAP

Draw a body outline and place red bone marrow, the thymus, lymph nodes, the spleen, the tonsils, and MALT where each one is found. Color the two primary organs one color and the four secondary organs another. Beside each, write its role in a few words.

Then: Draw arrows from the red bone marrow and the thymus to the secondary organs to show where mature lymphocytes go.

PASS 1 COLOR _____

PASS 2 COLOR _____

DRAW IT

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108
COMP

MODULE 4 · THE LYMPHATIC SYSTEM · LECTURE

Lymphatic disorders

You should be able to: Name edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis and identify the structure each one affects.

PICK ONE AND DO THAT ONE

☐ **A** DISORDER MAP

Draw a body outline with a tissue space, a lymphatic vessel in one leg, the nodes of the neck, the spleen, and the tonsils. Place each disorder at the structure it affects, with a tiny sketch of what changes: edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis.

Then: Circle the two disorders that are swelling from fluid, and write how edema differs from lymphedema.

☐ **B** DISORDER TABLE

Draw a table with edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis down the side. Across the top: the structure affected and what happens there. Use a word or a tiny sketch in each cell.

Then: Star the disorder that makes contact sports dangerous during mononucleosis and say which organ is at risk.

DRAW IT

split it up however the prompt needs

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109

COMP

MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Four-layer gut wall

You should be able to: Name the four layers of the alimentary canal wall from lumen outward and state the dominant tissue of each.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CROSS SECTION

Draw a cross section of the gut tube as a set of rings around the lumen. From the lumen outward, label the mucosa (epithelium, lamina propria, muscularis mucosae), the submucosa, the muscularis (inner circular and outer longitudinal layers), and the serosa. Write the main tissue of each layer inside its ring.

Then: Shade the layer that is also called the visceral peritoneum, and write what replaces it on the esophagus.

☐ **B** COMPARE TABLE

Draw a table with the four wall layers down the side, listed in order from the lumen outward. Across the top: the main tissue of the layer, and the smaller parts found inside it, such as vessels, a nerve plexus, or a thin muscle layer. Use a word or a tiny sketch in each cell.

Then: Write below the table how the outer layer of the esophagus and the muscularis of the stomach break the four-layer pattern.

DRAW IT

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Dr. Sharilyn Rennie

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110
COMP

MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Enteric nerve plexuses

You should be able to: Locate the submucosal and myenteric plexuses in the gut wall and match each to the wall layer it sits within.

PICK ONE AND DO THAT ONE

☐ **A** LABELED WALL WEDGE

Draw a wedge of gut wall as it looks on a slide, with the lumen at the bottom. Label the mucosa, submucosa, inner circular muscle, outer longitudinal muscle, and serosa. Draw the submucosal plexus as a dotted network inside the submucosa and the myenteric plexus as a dotted network between the two muscle layers.

Then: Write beside each plexus whether it controls secretion or movement, and name the system the two plexuses make together.

☐ **B** PLEXUS COMPARE TABLE

Draw a table with the submucosal plexus and the myenteric plexus down the side. Across the top: the wall layer it sits in, the layer just inside it (toward the lumen), the layer just outside it, and what it controls. Use a word or a tiny sketch in each cell.

Then: Write the name of the system the two plexuses form above the table, and circle the plexus that sits within the muscularis.

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MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Peritoneum and folds

You should be able to: Identify the parietal and visceral peritoneum, the peritoneal cavity, and the five folds greater omentum, lesser omentum, falciform ligament, mesentery, and mesocolon and state what each fold connects.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SAGITTAL VIEW

Draw a side view of the abdomen cut down the middle, with the diaphragm at the top, the anterior wall on one side, and the posterior wall on the other. Draw the liver, stomach, transverse colon, and coils of small intestine. Label the parietal peritoneum, visceral peritoneum, peritoneal cavity, greater omentum, lesser omentum, falciform ligament, mesentery, and mesocolon.

Then: Write beside each of the five folds the two things it connects, or what it hangs from and covers.

☐ **B** CONNECTION MAP

Draw boxes for the anterior abdominal wall, diaphragm, posterior abdominal wall, liver, stomach, duodenum, jejunum and ileum, transverse colon, sigmoid colon, and intestines. For each of the five folds, draw a line between the boxes it connects and write the fold's name on the line.

Then: Draw two curved lines below the map for the parietal and visceral peritoneum, and label the peritoneal cavity between them.

DRAW IT

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MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Retroperitoneal organs

You should be able to: Distinguish retroperitoneal organs from intraperitoneal ones and name the kidneys, pancreas, duodenum, and parts of the colon as retroperitoneal.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CROSS SECTION

Draw a cross section through the abdomen with the back at the bottom of your box. Draw the parietal peritoneum lining the posterior wall. Place the kidneys, pancreas, duodenum, and a retroperitoneal part of the colon behind it. Then draw a loop of jejunum hanging into the peritoneal cavity on a mesentery, wrapped in visceral peritoneum.

Then: Color every retroperitoneal organ the same color, and write beside the jejunum why it is not retroperitoneal.

☐ **B** SORTING BOX

Draw a box with two columns: behind the peritoneum, and inside the peritoneal cavity on a fold. Sort these organs: kidneys, pancreas, duodenum, ascending colon, descending colon, stomach, liver, jejunum, ileum, transverse colon, and sigmoid colon. Beside each organ in the second column, write the fold that holds it.

Then: Write across the top of the box, in your own words, what makes an organ retroperitoneal.

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COMP

MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Mouth, pharynx, esophagus

You should be able to: Identify the oral cavity boundaries, hard and soft palate, uvula, fauces, tonsils, the three pharyngeal regions, and the esophagus with its hiatus.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SAGITTAL VIEW

Draw a side view of the head and neck cut down the middle. Label the oral cavity, hard palate, soft palate, uvula, fauces, palatine tonsil, lingual tonsil, nasopharynx, oropharynx, laryngopharynx, and esophagus. Continue the esophagus down to a line for the diaphragm and label the esophageal hiatus.

Then: Write the boundaries of the oral cavity beside the mouth, and draw the uvula lifted to close off the nasopharynx.

☐ **B** FRONT VIEW AND CHAIN

Draw the open mouth from the front. Label the lips, cheeks, hard palate, soft palate, uvula, fauces, the palatine tonsils between the palatal arches, and the tongue with the lingual tonsil at its base. Beside it, draw boxes for the path of food from the fauces through the oropharynx, laryngopharynx, esophagus, and esophagus to the stomach, with a side box for the nasopharynx.

Then: Write beside your chain the two bones that form the hard palate and the length of the esophagus.

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COMP

MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Major canal sphincters

You should be able to: Locate the upper and lower esophageal, pyloric, ileocecal, and internal and external anal sphincters and state which regions each one separates.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CANAL OUTLINE

Draw a simple outline of the alimentary canal from the pharynx to the anus. Draw each sphincter as a ring and label it: upper esophageal, lower esophageal (cardiac), pyloric, ileocecal, internal anal, and external anal. Label the region on each side of every ring.

Then: Write the muscle type beside the pyloric, ileocecal, internal anal, and external anal sphincters, and color the one you control by choice.

☐ **B** SPHINCTER TABLE

Draw a table with the six sphincters down the side, in the order food meets them. Across the top: the region just before it, the region just after it, and any second name it has. Use a word or a tiny sketch in each cell.

Then: Star the two sphincters that guard the anus, and write which one you control by choice.

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COMP

MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Stomach regions and features

You should be able to: Identify the cardia, fundus, body, pyloric part, curvatures, rugae, and the three-layered muscularis of the stomach.

PICK ONE AND DO THAT ONE

☐ **A** LABELED ANTERIOR VIEW

Draw the J-shaped stomach below the diaphragm with the esophagus entering and the duodenum leaving. Label the cardia, fundus, body, pyloric part (pyloric antrum, pyloric canal, pylorus), pyloric sphincter, and the lesser and greater curvatures. Cut a window in the front wall to show the rugae, and peel back the muscle layers at one edge.

Then: Label the three muscle layers in order from inside to outside, and write which layer the rest of the tube does not have.

☐ **B** REGION TABLE

Draw a table with the cardia, fundus, body, pyloric part, lesser curvature, greater curvature, and rugae down the side. Across the top: where it is on the stomach and what it looks like. Use a word or a tiny sketch in each cell.

Then: Draw a strip of stomach wall below the table, and label its inner oblique, middle circular, and outer longitudinal muscle layers.

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MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Small intestine and surface area

You should be able to: Identify the duodenum, jejunum, and ileum and name the circular folds, villi, lacteal, and microvilli as features that expand absorptive surface area.

PICK ONE AND DO THAT ONE

☐ **A** LABELED MODEL VIEW

Draw the small intestine as it sits in the abdomen, from the pyloric sphincter to the ileocecal sphincter. Label the duodenum, jejunum, and ileum with the length of each, and the mesentery on the jejunum and ileum. Off the jejunum, draw a zoom bubble showing a circular fold, one villus with its lacteal, and microvilli on one absorptive cell.

Then: Shade the region that is mostly retroperitoneal, and number the three surface features from largest to smallest.

☐ **B** ZOOM SEQUENCE

Draw a straight strip divided into the duodenum, jejunum, and ileum, with the length of each written on it. From one spot, draw three zooms, each smaller than the last: a circular fold of mucosa and submucosa, one villus with its lacteal, and one absorptive cell with microvilli forming the brush border.

Then: Write beside the lacteal what it absorbs, and beside the ileum the sphincter where it joins the large intestine.

DRAW IT

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COMP

MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Large intestine features

You should be able to: Identify the cecum, appendix, four parts of the colon, rectum, and anal canal along with the teniae coli, haustra, and omental appendices.

PICK ONE AND DO THAT ONE

☐ **A** LABELED ANTERIOR VIEW

Draw the large intestine framing the abdomen, with the ileum entering at the ileocecal valve. Label the cecum, vermiform appendix, ascending colon, transverse colon, descending colon, sigmoid colon, rectum, anal canal, and the internal and external anal sphincters. On the transverse colon, draw and label the teniae coli, haustra, and omental appendices.

Then: Write beside the rectum and the anal canal where each one sits, and color the two colon parts that hang on a mesocolon.

☐ **B** REGION TABLE

Draw a table with the cecum, the four parts of the colon, the rectum, and the anal canal down the side, in the order feces move through them. Across the top: what comes just before it, what comes just after it, and one thing that helps you find it on the model. Use a word or a tiny sketch in each cell.

Then: Draw a cecum with its appendix below the table, and a short piece of colon with three teniae coli, haustra, and omental appendices.

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COMP

MODULE 4 · THE ALIMENTARY CANAL · LECTURE AND LAB

Trace the pathway of food

You should be able to: Trace a bite of food through every named segment of the alimentary canal in order from mouth to anus.

PICK ONE AND DO THAT ONE

☐ **A** FLOW CHAIN

Draw a chain of boxes with arrows that carries a bite of food from the mouth to the anus. Put every named segment in its own box, including the two parts of the pharynx that food passes, the three parts of the small intestine, and each part of the large intestine. Draw a ring on the arrow wherever a sphincter sits.

Then: Write one word under each box for what happens to the food there, using the pathway in your notes.

☐ **B** BODY OUTLINE MAP

Draw an outline of the head and trunk with the diaphragm. Draw the whole canal in place, from the mouth through the pharynx, esophagus, stomach, small intestine, and large intestine to the anus. Number every named segment in the order food reaches it, and label each number.

Then: Circle the region where most absorption happens, and box the region where water is absorbed and feces form.

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COMP

MODULE 4 · THE ALIMENTARY CANAL · LECTURE

Clinical anatomy of canal disorders

You should be able to: Match GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, IBD, hemorrhoids, and colorectal cancer to the canal structure each one affects.

PICK ONE AND DO THAT ONE

☐ **A** DISORDER MAP

Draw an outline of the canal from the esophagus to the anus, with the diaphragm and its esophageal hiatus. Write each disorder on the structure it affects: GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, inflammatory bowel disease, hemorrhoids, and colorectal cancer. Draw a small sketch of the problem beside each label, such as a pouch or a swollen vein.

Then: Write beside the lower esophageal sphincter and the appendix where the patient feels the pain from each problem.

☐ **B** COMPARE TABLE

Draw a table with the eight disorders down the side: GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, inflammatory bowel disease, hemorrhoids, and colorectal cancer. Across the top: the structure it affects and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Draw the four wall layers beside the table, and show how a peptic ulcer erodes from the mucosa until it perforates into the peritoneal cavity.

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COMP

MODULE 4 · THE ACCESSORY DIGESTIVE ORGANS · LECTURE AND LAB

Teeth and tongue

You should be able to: Identify the teeth and tongue with the lingual frenulum, papillae, intrinsic and extrinsic muscles, and lingual glands.

PICK ONE AND DO THAT ONE

☐ **A** LABELED MOUTH VIEW

Draw the open mouth from the front with the upper and lower teeth and the tongue lifted toward the roof of the mouth. Label the teeth, the tongue, and the lingual frenulum under it. Beside it, draw a side view of the tongue and label the papillae on its upper surface, the intrinsic muscles, the extrinsic muscles, and the lingual glands.

Then: Draw arrows for the movements the extrinsic muscles make, and write what the intrinsic muscles change about the tongue.

☐ **B** STRUCTURE TABLE

Draw a table with the teeth, tongue, lingual frenulum, papillae, intrinsic tongue muscles, extrinsic tongue muscles, and lingual glands down the side. Across the top: where it is and what it does. Use a word or a tiny sketch in each cell.

Then: Circle the two muscle groups, and write beside them which group starts outside the tongue and which sits within it.

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COMP

MODULE 4 · THE ACCESSORY DIGESTIVE ORGANS · LECTURE AND LAB

Salivary glands

You should be able to: Locate the parotid, submandibular, and sublingual glands and state where the duct of each opens.

PICK ONE AND DO THAT ONE

☐ **A** LABELED FACE VIEW

Draw a side view of the face with the ear, the mandible, the masseter muscle, and the inside of the mouth. Draw and label the parotid, submandibular, and sublingual glands in place. Draw each gland's duct running to the spot where it opens inside the mouth, and label that spot.

Then: Color each gland and its duct opening the same color, and write the muscle the parotid gland lies over.

☐ **B** LOCATION TABLE

Draw a table with the parotid, submandibular, and sublingual glands down the side. Across the top: where the gland sits, the landmark next to it, and where its duct opens in the mouth. Use a word or a tiny sketch in each cell.

Then: Sketch the floor of the mouth below the table with the tongue lifted, and mark where the submandibular and sublingual ducts open.

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MODULE 4 · THE ACCESSORY DIGESTIVE ORGANS · LECTURE AND LAB

Liver gross and lobule anatomy

You should be able to: Identify the four liver lobes, falciform ligament, and ligamentum teres, and name the hepatocytes, sinusoids, canaliculi, and portal triad of the lobule.

PICK ONE AND DO THAT ONE

☐ **A** LABELED VIEWS AND ZOOM

Draw the liver from the front and label the right lobe, left lobe, falciform ligament, and ligamentum teres. Draw a small view of its posterior and inferior surface with the quadrate and caudate lobes. Add a zoom bubble of one lobule with plates of hepatocytes, sinusoids between the plates, bile canaliculi, and a portal triad at one corner.

Then: Label the three parts of the portal triad, and write the fetal vessel the ligamentum teres is a remnant of.

☐ **B** GROSS TO MICRO TABLE

Draw a table with the right, left, quadrate, and caudate lobes, the falciform ligament, ligamentum teres, hepatocytes, sinusoids, bile canaliculi, and portal triad down the side. Across the top: seen by eye or on a slide, where it is, and what it is or does. Use a word or a tiny sketch in each cell.

Then: Draw one portal triad below the table, and add arrows for blood coming in and bile going out.

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MODULE 4 · THE ACCESSORY DIGESTIVE ORGANS · LECTURE AND LAB

Gallbladder and bile pathway

You should be able to: Identify the gallbladder regions and trace bile from the canaliculi through the hepatic, cystic, and common bile ducts to the hepatopancreatic ampulla and duodenum.

PICK ONE AND DO THAT ONE

☐ **A** LABELED BILIARY TREE

Draw the underside of the liver with the gallbladder, the duodenum below, and the pancreas beside it. Add a small bubble of bile canaliculi between hepatocytes. Label the fundus, body, and neck of the gallbladder, the right and left hepatic ducts, common hepatic duct, cystic duct, common bile duct, pancreatic duct, hepatopancreatic ampulla, sphincter of Oddi, and major duodenal papilla.

Then: Draw arrows that carry bile from the canaliculi to the duodenum, with a side loop through the cystic duct into the gallbladder.

☐ **B** FLOW CHAIN

Draw a chain of boxes with arrows that carries a drop of bile from the hepatocyte that made it to the duodenum. Give every duct and opening its own box, starting at the bile canaliculi. Branch a side arrow off the chain to the gallbladder, and sketch the gallbladder with its fundus, body, and neck.

Then: Write on the chain where the pancreatic duct joins and where the sphincter of Oddi controls flow.

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MODULE 4 · THE ACCESSORY DIGESTIVE ORGANS · LECTURE AND LAB

Pancreas structure

You should be able to: Identify the head, body, and tail of the pancreas with its main and accessory ducts, major duodenal papilla, sphincter of Oddi, acini, and islets.

PICK ONE AND DO THAT ONE

☐ **A** LABELED VIEW AND ZOOM

Draw the pancreas behind the stomach with its head in the curve of the duodenum. Label the head, body, and tail, the main pancreatic duct, the accessory duct, the common bile duct joining at the hepatopancreatic ampulla, the sphincter of Oddi, and the major duodenal papilla. Add a zoom bubble with acini and one pancreatic islet.

Then: Write the second name of each duct, and label which cluster is exocrine and which is endocrine, with its percent of the gland.

☐ **B** COMPARE TABLE

Draw a table with the head, body, tail, main pancreatic duct, accessory duct, major duodenal papilla, sphincter of Oddi, acini, and pancreatic islets down the side. Across the top: where it is, its other name if it has one, and what it does or carries. Use a word or a tiny sketch in each cell.

Then: Sketch the two ducts entering the duodenum below the table, and circle the one that joins the common bile duct.

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MODULE 4 · THE ACCESSORY DIGESTIVE ORGANS · LECTURE

Clinical anatomy of accessory disorders

You should be able to: Match gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps to the accessory organ each one affects and explain how a stone at the ampulla can block bile and pancreatic flow at once.

PICK ONE AND DO THAT ONE

☐ **A** DISORDER MAP

Draw the liver, gallbladder, bile ducts, pancreas, and duodenum together, and a small side view of the face with the parotid gland. Write each disorder on the organ it affects: gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps. Draw a gallstone stuck at the hepatopancreatic ampulla.

Then: Draw arrows showing bile and pancreatic juice both backed up behind the stone, and write which disorder this can trigger.

☐ **B** COMPARE TABLE

Draw a table with gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps down the side. Across the top: the organ it affects and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Draw the bile duct and pancreatic duct meeting at the ampulla below the table, place a stone there, and write which two flows stop.

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126

COMP

MODULE 4 · LAB: VESSELS AND NERVES OF THE ABDOMEN AND LOWER LIMB · LAB

Abdominal aorta and portal system

You should be able to: Identify the major branches of the abdominal aorta and the vessels of the hepatic portal system on the cadaver.

PICK ONE AND DO THAT ONE

☐ **A** LABELED ANTERIOR VIEW

Draw the abdominal aorta and inferior vena cava. Label the celiac trunk with its left gastric, splenic and common hepatic arteries, then the superior mesenteric, renal, gonadal and inferior mesenteric arteries, and the right and left common iliacs. Above, label the liver, hepatic veins, hepatic portal vein, the splenic and superior mesenteric veins that form it, and the inferior mesenteric vein joining the splenic vein.

Then: Number the three unpaired aortic branches 1 to 3 from top to bottom, and write beside the liver what makes this a portal system.

☐ **B** BRANCH AND DRAINAGE TABLE

Draw a table with these vessels down the side: celiac trunk, left gastric, splenic, common hepatic, superior mesenteric, renal, gonadal and inferior mesenteric arteries, then the hepatic portal, splenic, superior mesenteric, inferior mesenteric and hepatic veins. Across the top: paired or unpaired, and the vessel it branches off or drains into. Use a word in each cell.

Then: Below the table, write the chain of vessels a glucose molecule follows from an intestinal capillary to the right atrium, and circle the two capillary beds.

DRAW IT

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COMP

MODULE 4 · LAB: VESSELS AND NERVES OF THE ABDOMEN AND LOWER LIMB · LAB

Lower limb vessels and nerves

You should be able to: Identify the femoral and lower-limb blood vessels and the major nerves of the abdomen and lower limb on the cadaver.

PICK ONE AND DO THAT ONE

☐ **A** FRONT AND BACK VIEWS

Outline the right lower limb twice, from the front and from the back. On the front view, label the femoral artery, femoral vein, femoral nerve and great saphenous vein in the thigh, and carry the great saphenous vein down the leg. On the back view, label the sciatic nerve, its tibial and common fibular branches, and the posterior tibial artery in the leg.

Then: Write deep or superficial beside each vein, and write lumbar plexus or sacral plexus beside each nerve.

☐ **B** STRUCTURE-TO-REGION TABLE

Draw a table with the femoral artery, femoral vein, great saphenous vein, posterior tibial artery, femoral nerve, sciatic nerve, tibial nerve and common fibular nerve down the side. Across the top: artery, vein or nerve; found in the thigh, the leg, or both; and deep or superficial (veins only). Use a word or a tiny sketch in each cell.

Then: Below the table, draw two boxes for the lumbar and sacral plexuses and branch each nerve from the plexus or nerve it comes from.

DRAW IT

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The box is **drawn**, not written. Pick one prompt and divide the box however it needs. Words go inside the drawing, in labels and little boxes with arrows between them. Sentences running across the page do not count.

128
COMP

MODULE 4 · THE ENDOCRINE SYSTEM · LECTURE

Major endocrine glands and locations

You should be able to: Name the major endocrine glands and locate each one in the body from the hypothalamus to the gonads.

PICK ONE AND DO THAT ONE

☐ **A** LABELED BODY MAP

Draw a body outline from the head to the pelvis. Place and label the hypothalamus, pituitary gland, pineal gland, thyroid gland, parathyroid glands, thymus, adrenal glands, pancreatic islets, ovaries and testes. Beside each one, write its landmark from your notes, such as the third ventricle, the sella turcica, the larynx, the sternum, the kidney, the stomach, the uterus or the scrotum.

Then: Star the largest purely endocrine gland, and write beside the thymus how its size changes after puberty.

☐ **B** LOCATION TABLE

Draw a table with the ten major endocrine glands down the side, from the hypothalamus to the testes. Across the top: the body region (head, neck, chest, abdomen, pelvis or scrotum), and the bone, organ or space it sits in or against. Use a word or a tiny sketch in each cell.

Then: Circle every gland that comes in a pair or more, and write beside the parathyroid glands how many there usually are.

DRAW IT

split it up however the prompt needs

PASS 1 COLOR _____

PASS 2 COLOR _____

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Dr. Sharilyn Rennie

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COMP

MODULE 4 · THE ENDOCRINE SYSTEM · LECTURE AND LAB

Pituitary gross anatomy and lobes

You should be able to: Describe the pituitary in the sella turcica, its infundibulum link to the hypothalamus, and compare the anterior and posterior lobes by tissue type.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SAGITTAL VIEW

Draw a midsagittal view of the base of the brain sitting over the sphenoid bone. Label the hypothalamus, infundibulum, sella turcica, and the pituitary gland (hypophysis) with its anterior lobe (adenohypophysis) and posterior lobe (neurohypophysis). Draw the nerve fibers running down the infundibulum to the posterior lobe and the hypophyseal portal vessels reaching the anterior lobe.

Then: Write the tissue type inside each lobe, then add an arrow showing which way a pituitary tumor grows and label what it presses on.

☐ **B** TWO-LOBE COMPARE TABLE

Draw a table with the anterior lobe and the posterior lobe down the side. Across the top: its other name, the tissue it is made of, and how it links to the hypothalamus. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch the gland in the sella turcica hanging from the infundibulum, and write one sentence on how the tissue type predicts the link.

DRAW IT

split it up however the prompt needs

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PASS 2 COLOR _____

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COMP

MODULE 4 · THE ENDOCRINE SYSTEM · LECTURE AND LAB

Thyroid and parathyroid anatomy

You should be able to: Identify the thyroid lobes, isthmus, follicles, colloid, follicular and parafollicular cells, and the parathyroid chief and oxyphil cells.

PICK ONE AND DO THAT ONE

☐ **A** LABELED VIEWS WITH INSETS

Draw the thyroid wrapped on the trachea just below the larynx, seen from the front, and label the right lobe, left lobe, isthmus, trachea and larynx. Beside it, draw a back view with the usual four parathyroid glands. Add one magnified circle of thyroid follicles labeling colloid, follicular cells and parafollicular (C) cells, and a second circle of parathyroid tissue labeling chief cells.

Then: Write beside the back view why the parathyroid glands are at risk during thyroid surgery.

☐ **B** ZOOM-IN SEQUENCE

Draw three boxes in a row, each zooming in on the one before: the neck with the thyroid on the trachea, the gland with its lobes and isthmus, then a group of follicles. In the last box, label colloid, follicular cells in the wall, and parafollicular (C) cells between follicles. Add a side box for a parathyroid gland and label its chief cells.

Then: Write under each box whether you would identify it on the model, the cadaver, or the slide.

DRAW IT

split it up however the prompt needs

PASS 1 COLOR _____

PASS 2 COLOR _____

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Dr. Sharilyn Rennie

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COMP

MODULE 4 · THE ENDOCRINE SYSTEM · LECTURE AND LAB

Adrenal gland zones

You should be able to: Identify the adrenal cortex zones in order and the adrenal medulla with its chromaffin cells.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CUTAWAY

Draw a kidney with the adrenal gland capping its superior pole, then cut a wedge out of the adrenal to show its inside. Label the capsule, the adrenal cortex with its three zones in order from the surface inward (zona glomerulosa, zona fasciculata, zona reticularis), and the adrenal medulla at the core. Draw each zone at its relative thickness.

Then: Write the tissue type beside the cortex and beside the medulla, and sketch the cell pattern of the fasciculata and the reticularis.

☐ **B** LAYER TABLE

Draw a table with the zona glomerulosa, zona fasciculata, zona reticularis and adrenal medulla down the side, in order from the surface inward. Across the top: its position (outer, middle, inner or core), its thickness, and its tissue type. Use a word or a tiny sketch in each cell.

Then: Below the table, write why imaging ordered for a kidney often shows the adrenal gland as well.

DRAW IT

split it up however the prompt needs

PASS 1 COLOR _____

PASS 2 COLOR _____

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COMP

MODULE 4 · THE ENDOCRINE SYSTEM · LECTURE

Pancreas, gonads, and organs with endocrine tissue

You should be able to: Locate the pancreatic islets and gonads and name other organs such as the heart, kidneys, and skin that contain endocrine tissue.

PICK ONE AND DO THAT ONE

☐ **A** LABELED BODY MAP

Draw a body outline. Draw the pancreas behind the stomach with a magnified circle of pancreatic islets among the acini, the ovaries beside the uterus, and the testes in the scrotum. Then star the heart, kidneys, skin, stomach and small intestine, liver, adipose tissue and placenta, and label the endocrine tissue each one contains, such as cells in the wall of the atria.

Then: Write endocrine on the islets and exocrine on the acini, and write beside the placenta what makes it a temporary organ.

☐ **B** ORGAN SORTING TABLE

Draw a table with the pancreas, ovaries, testes, heart, kidneys, skin, stomach and small intestine, liver, adipose tissue and placenta down the side. Across the top: where it sits, and the endocrine tissue it contains. Use a word or a tiny sketch in each cell.

Then: Circle the three organs your notes call mixed organs, and write beside them what makes an organ mixed.

DRAW IT

split it up however the prompt needs

PASS 1 COLOR _____

PASS 2 COLOR _____

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Dr. Sharilyn Rennie

PART C

Check your drawing.

Do this after both passes. Find your competency and the prompt you picked, and tick off everything on the list that is in your box. Anything missing goes in with your second color. Then look for the common mistake named underneath.

97 Upper and lower tracts, conducting and respiratory zones

PROMPT A · LABELED AIRWAY OUTLINE

- ☐ Upper tract bracket covers the nose, nasal cavity, paranasal sinuses and pharynx
- ☐ Lower tract bracket covers the larynx, trachea, bronchi and lungs
- ☐ Conducting zone bracket runs from the nose all the way down to the terminal bronchioles
- ☐ Respiratory zone bracket covers only the respiratory bronchioles, alveolar ducts and alveoli, deep in the lungs
- ☐ The larynx, trachea and bronchi sit in the lower tract but still in the conducting zone, so the two sets of brackets do not line up
- ☐ Zoom box runs in order: terminal bronchiole, respiratory bronchiole, alveolar duct, alveoli
- ☐ Circle sits at the line between the terminal bronchiole and the respiratory bronchiole
- ☐ Terminal bronchiole named as the last conducting airway, respiratory bronchiole as the first airway of the respiratory zone

Check: The conducting zone ends at the terminal bronchioles, not at the bronchi or where the lung begins. Check where your circle sits.

PROMPT B · SORTING GRID

- ☐ Upper tract, conducting zone box: nose, nasal cavity, paranasal sinuses and pharynx
- ☐ Lower tract, conducting zone box: larynx, trachea, bronchi, bronchioles and terminal bronchioles
- ☐ Lower tract, respiratory zone box: respiratory bronchioles, alveolar ducts and alveoli
- ☐ Upper tract, respiratory zone box is the empty one
- ☐ Reason written in it: gas exchange happens only deep in the lungs, and every upper tract structure only carries air
- ☐ Three jobs of the conducting zone: it warms, moistens and filters the air
- ☐ Sketches match the structure, such as a ringed tube for the trachea and tiny round sacs for the alveoli

Check: Terminal bronchioles go in the conducting zone. Only the respiratory bronchioles and the airways past them exchange gas.

98 Nose, nasal cavity, and pharynx

PROMPT A · LABELED SAGITTAL VIEW

- ☐ Nares open into the nasal vestibule just inside the nose
- ☐ Superior, middle and inferior conchae drawn as shelves on the side wall, with a meatus as the passage under each one
- ☐ Hard palate in front and soft palate behind form the floor of the nasal cavity
- ☐ Choanae at the back of the nasal cavity, where it opens into the nasopharynx
- ☐ Frontal sinus in the forehead above the nasal cavity, sphenoidal sinus in the sphenoid bone, deep in the skull behind the nasal cavity
- ☐ Nasopharynx above the soft palate, oropharynx from the soft palate to the epiglottis, laryngopharynx from the epiglottis to the esophagus
- ☐ Pharyngeal tonsil on the back wall of the nasopharynx, auditory tube opening on its side wall
- ☐ Palatine tonsil in the oropharynx and lingual tonsil at the base of the tongue
- ☐ Front view: frontal sinuses above the eyes, ethmoid air cells between the eyes, maxillary sinuses in the cheeks below the eyes
- ☐ Sphenoidal sinus named as the one hidden behind the ethmoid air cells

Check: A meatus is the passage under a concha, not the concha itself. Make sure each meatus label points to a space.

PROMPT B · FLOW CHAIN

- ☐ Boxes in order: nares, nasal vestibule, nasal cavity, choanae, nasopharynx, oropharynx, laryngopharynx
- ☐ Nasal septum split down the middle: septal cartilage in front, vomer and perpendicular plate of the ethmoid behind
- ☐ Three conchae on the side wall with a meatus under each one
- ☐ Hard palate in front and soft palate behind drawn as the floor
- ☐ Frontal, ethmoid, sphenoid and maxillary sinuses all branch off the nasal cavity box, because they drain into it
- ☐ Nasopharynx: air only, pseudostratified ciliated columnar, holds the pharyngeal tonsil
- ☐ Oropharynx: air and food, stratified squamous, holds the palatine and lingual tonsils
- ☐ Laryngopharynx: air and food, stratified squamous, no tonsil

Check: Only the nasopharynx carries air alone and keeps the ciliated lining. The two lower regions carry food too and switch to stratified squamous.

99 Larynx

PROMPT A · TWO LABELED VIEWS

- [ ] Hyoid bone at the top, above the larynx
- [ ] Epiglottis showing above the thyroid cartilage
- [ ] Thyroid cartilage as the large shield, with the laryngeal prominence at the front midline
- [ ] Cricothyroid ligament filling the gap between the thyroid and cricoid cartilages
- [ ] Cricoid cartilage as the ring just below the thyroid cartilage, sitting on top of the trachea
- [ ] Superior view: epiglottis at the front, vestibular folds above and outside the pale vocal folds
- [ ] Glottis labeled as the vocal folds plus the opening between them
- [ ] Arytenoid cartilages at the back where the vocal folds attach, corniculates on top of them, cuneiforms in the fold running to the epiglottis
- [ ] Epiglottis colored as elastic cartilage
- [ ] Star on the cricoid cartilage as the only complete ring

Check: The star belongs on the cricoid, not the thyroid cartilage. The thyroid cartilage is a shield that is open at the back.

PROMPT B · CARTILAGE AND FOLD TABLE

- [ ] Thyroid cartilage: single, large shield forming the laryngeal prominence (Adam's apple)
- [ ] Cricoid cartilage: single, a complete ring just below the thyroid cartilage
- [ ] Epiglottis: single, spoon-shaped flap that folds over the opening of the larynx when you swallow
- [ ] Arytenoid cartilages: paired, at the back, anchor and move the vocal folds
- [ ] Corniculate cartilages: paired, sit on top of the arytenoids
- [ ] Cuneiform cartilages: paired, in the fold between the arytenoid and the epiglottis
- [ ] Vocal folds: fold, the true vocal cords, pearly white, the lower pair
- [ ] Vestibular folds: fold, the false vocal cords, above the true folds
- [ ] Glottis: the vocal folds together with the opening between them
- [ ] Vocal ligament runs inside the vocal fold from the arytenoid at the back to the thyroid cartilage at the front, and only the vocal folds are marked as making sound

Check: The vestibular folds are the false cords and make no sound. Only the true vocal folds get the sound mark.

100 Trachea and bronchial tree

PROMPT A · LABELED BRONCHIAL TREE

- [ ] Tracheal cartilages drawn as C shapes, open at the back
- [ ] Carina at the fork where the trachea splits into the two main bronchi
- [ ] Right main bronchus drawn wider and more vertical than the left
- [ ] Lobar bronchi: three on the right and two on the left, one to each lobe
- [ ] Segmental bronchi branching off the lobar bronchi, one to each bronchopulmonary segment
- [ ] Bronchioles past the segmental bronchi, ending in terminal bronchioles
- [ ] Numbers run in order: trachea, carina, main bronchi, lobar bronchi, segmental bronchi, bronchioles, terminal bronchioles
- [ ] Cross section: C ring open toward the back with the trachealis muscle closing the gap
- [ ] Esophagus drawn right behind the trachealis muscle

Check: Lobar bronchi match the lobes, so there are three on the right and two on the left. Count yours.

PROMPT B · FLOW CHAIN

- [ ] Boxes in order: trachea, carina, main bronchi, lobar bronchi, segmental bronchi, bronchioles, terminal bronchioles
- [ ] Main bronchi are the primary bronchi, one to each lung
- [ ] Lobar bronchi are the secondary bronchi, one to each lobe
- [ ] Segmental bronchi are the tertiary bronchi, one to each bronchopulmonary segment
- [ ] Trachea wall sketch: C ring of hyaline cartilage with the trachealis muscle at the back
- [ ] Bronchiole wall sketch: a ring of smooth muscle and no cartilage
- [ ] Terminal bronchioles noted as the last airway of the conducting zone
- [ ] Bar shows cartilage shrinking and smooth muscle growing as you move down the chain
- [ ] Mark for cartilage disappearing sits at the bronchioles
- [ ] Lobar and segmental bronchi wall sketch: irregular plates of cartilage with a more complete layer of smooth muscle

Check: Cartilage is already gone at the bronchioles, not just at the terminal bronchioles. Check where your mark sits.

101 Lungs and pleurae

PROMPT A · LABELED LUNG VIEWS

- [ ] Right lung has three lobes: superior, middle and inferior
- [ ] Horizontal fissure between the right superior and middle lobes
- [ ] Oblique fissure on each lung, cutting off the inferior lobe
- [ ] Left lung has two lobes: superior and inferior, with no horizontal fissure
- [ ] Cardiac notch cut into the front edge of the left lung where the heart sits
- [ ] Lingula as the tongue-shaped piece of the left superior lobe, just below the cardiac notch
- [ ] Apex at the narrow top of each lung, base as the broad bottom resting on the diaphragm
- [ ] Hilum on the medial surface of each lung, where the bronchus and vessels enter
- [ ] Parietal pleura as the outer line lining the chest wall, visceral pleura as the inner line on the lung surface
- [ ] Pleural cavity as the thin, fluid-filled gap between the two pleurae

Check: The left lung has two lobes and only an oblique fissure. Count your lobes and fissures.

PROMPT B · COMPARE TABLE

- [ ] Number of lobes: right three, left two
- [ ] Lobe names: right superior, middle and inferior; left superior and inferior
- [ ] Fissures: right has a horizontal and an oblique fissure, left has an oblique fissure only
- [ ] Special features: the left lung has the cardiac notch and the lingula
- [ ] Size: the right lung is larger, the left is smaller to make room for the heart
- [ ] Parietal pleura as the outer layer lining the chest wall
- [ ] Visceral pleura as the inner layer on the surface of the lung
- [ ] Pleural cavity as the thin, fluid-filled space between the two layers
- [ ] Apex at the top, base at the bottom on the diaphragm, hilum on the medial surface

Check: The cardiac notch and the lingula belong to the left lung only. Check that your right lung column does not list them.

102 Respiratory zone and alveolar structure

PROMPT A · LABELED ALVEOLAR CLOSE-UP

- [ ] Respiratory bronchiole with a few alveoli budding from its wall
- [ ] Alveolar duct with walls lined entirely by alveoli
- [ ] Alveolar sac as a cluster of alveoli sharing one space
- [ ] Pulmonary capillaries drawn as a dense net wrapped around the alveoli
- [ ] Type I alveolar cells as thin, flat cells making up most of the wall
- [ ] Type II alveolar cell as a cuboidal cell tucked into the wall, the one that makes surfactant
- [ ] Alveolar macrophage on the inner air side of the wall
- [ ] Pulmonary capillary pressed against the outside of the wall
- [ ] Membrane layers in order from air to blood: alveolar epithelium, fused basement membranes, capillary endothelium

Check: Type I cells are thin and flat and make up most of the wall. If your wall is mostly cuboidal cells, you swapped the two types.

PROMPT B · CELL AND STRUCTURE TABLE

- [ ] Type I alveolar cells: thin, flat simple squamous cells forming most of the alveolar wall, where gases cross
- [ ] Type II alveolar cells: cuboidal cells in the wall that secrete surfactant, which keeps the alveoli from collapsing
- [ ] Alveolar macrophages: dust cells on the inner surface of the alveolus that engulf debris
- [ ] Respiratory membrane: alveolar epithelium, fused basement membranes and capillary endothelium, the thin barrier between air and blood
- [ ] Pulmonary capillaries: a dense net wrapped around each alveolus, bringing blood close to the air
- [ ] Alveolus sketch shows the macrophage inside the air space and the capillaries on the outside
- [ ] Respiratory membrane placed where the alveolar wall touches a capillary

Check: Surfactant comes from the type II cells, not the type I cells. Check your job column.

103 Thoracic wall and diaphragm

PROMPT A · LABELED THORAX VIEW

- [ ] Ribs with the external intercostals as the outer layer between them and the internal intercostals deep to them
- [ ] Mediastinum in the middle of the chest, between the two lungs
- [ ] Diaphragm drawn as a dome separating the thoracic cavity above from the abdominal cavity below
- [ ] Inferior vena cava passes through the diaphragm at T8
- [ ] Esophagus passes through at T10
- [ ] Aorta passes through at T12
- [ ] External intercostal lines slant down and toward the front, like hands sliding into pockets
- [ ] Internal intercostal lines run at right angles to the external ones, slanting down and toward the back, the same slant as praying hands

Check: T8, T10 and T12 are vertebral levels, with the vena cava highest and the aorta lowest. Check that you did not flip the order.

PROMPT B · VERTEBRAL LEVEL MAP

- [ ] Diaphragm arches in front of the vertebral column
- [ ] Inferior vena cava passes through the diaphragm in line with T8
- [ ] Esophagus passes through in line with T10
- [ ] Aorta passes through in line with T12
- [ ] From top to bottom the order is vena cava, esophagus, aorta
- [ ] External intercostals as the outer layer between the two ribs
- [ ] Internal intercostals as the deeper layer, just inside the external ones
- [ ] Thoracic cavity labeled above the diaphragm and abdominal cavity below it

Check: T8, T10 and T12 are vertebral levels, with the vena cava highest and the aorta lowest. Check that you did not flip the order.

104 Respiratory disorders by structure affected

PROMPT A · DISORDER MAP

- [ ] Asthma on the bronchi and bronchioles, drawn inflamed and narrowed
- [ ] COPD on the airways and alveoli, a long-term blockage of airflow
- [ ] Emphysema on the alveoli, with the walls destroyed so small sacs merge into a few large ones
- [ ] Pneumonia on the alveoli, inflamed and filled with fluid
- [ ] Pneumothorax in the pleural cavity, with air in the space and the lung collapsed
- [ ] Lung cancer in the lung tissue, drawn as a tumor
- [ ] Inhaled object drawn in the right main bronchus
- [ ] Reason: the right main bronchus is wider, shorter and more vertical than the left

Check: Pneumothorax is air in the pleural cavity, not inside the lung. Your sketch should show air between the two pleurae.

PROMPT B · DISORDER TABLE

- [ ] Asthma: bronchi and bronchioles; the lining is inflamed, the smooth muscle tightens and extra mucus collects, so the airways narrow
- [ ] COPD: airways and alveoli, with long-term blockage of airflow; it includes emphysema and chronic bronchitis
- [ ] Emphysema: alveoli, whose walls are destroyed, shrinking the surface for gas exchange
- [ ] Pneumonia: alveoli, which are infected, inflamed and filled with fluid
- [ ] Pneumothorax: pleural cavity, where air gets in, the seal breaks and the lung collapses
- [ ] Lung cancer: lung tissue, with a malignant tumor strongly linked to smoking
- [ ] Star on emphysema
- [ ] Before sketch: many small alveoli. After sketch: a few large, merged sacs with less wall surface

Check: Pneumonia fills the alveoli with fluid but does not destroy them. The star goes on emphysema.

105 Lymph pathway

PROMPT A · FLOW MAP

- [ ] Fluid drawn leaking out of the blood capillary into the spaces between the cells
- [ ] Interstitial fluid labeled in the tissue spaces
- [ ] Lymphatic capillary sitting among the cells, taking up that fluid
- [ ] Lymph labeled inside the lymphatic capillary
- [ ] Fluid becomes lymph the moment it enters the lymphatic capillary
- [ ] Chain in order: lymphatic capillary, collecting vessel, lymph node, trunk, duct, subclavian vein
- [ ] Arrows point one way only, from the tissues toward the subclavian vein

Check: The fluid is called lymph only after it enters a lymphatic vessel. In the tissue spaces it is still interstitial fluid.

PROMPT B · DRAINAGE BODY MAP

- [ ] Path starts in a lymphatic capillary in the left foot
- [ ] Collecting vessel runs up the leg with valves drawn inside it
- [ ] Inguinal node in the groin
- [ ] Lumbar trunk running up the back wall of the abdomen to the cisterna chyli
- [ ] Cisterna chyli as the sac in the abdomen where the thoracic duct starts
- [ ] Thoracic duct runs up through the thorax and empties where the left subclavian and left internal jugular veins meet
- [ ] Shaded area is the right arm and the right side of the head, neck and thorax
- [ ] Arrowheads all point toward the heart, with none pointing back
- [ ] Definition: lymph is interstitial fluid once it has entered a lymphatic vessel

Check: The right lymphatic duct drains only the upper right part of the body. Your right leg should not be shaded.

106 Vessel and node structure

PROMPT A · LABELED NODE

- [ ] Lymphatic capillary closed at one end
- [ ] Overlapping endothelial cells drawn as one-way flap valves
- [ ] Anchoring filaments tying the capillary to the tissue around it
- [ ] Collecting vessel cut open with valves inside
- [ ] Capsule around the node with trabeculae reaching inward
- [ ] Subcapsular sinus just under the capsule
- [ ] Cortex with follicles near the surface, medulla with medullary cords and sinuses toward the center
- [ ] Several afferent vessels entering through the capsule and fewer efferent vessels leaving at the indented hilum
- [ ] Arrows run afferent vessels, subcapsular sinus, cortical sinuses, medullary sinuses, efferent vessels
- [ ] Lymph slows because more afferent vessels bring it in than efferent vessels take it out

Check: Afferent vessels enter all around the capsule, but efferent vessels leave only at the hilum. Draw more coming in than going out.

PROMPT B · COMPARE TABLE

- [ ] Lymphatic capillary wall: one layer of overlapping endothelial cells, held open by anchoring filaments
- [ ] Capillary keeps flow one way with its overlapping cells, which act as flap valves
- [ ] Capillary sends lymph next to a collecting vessel
- [ ] Collecting vessel wall: built like a vein, with three thin layers
- [ ] Collecting vessel keeps flow one way with many valves, more than veins have
- [ ] Collecting vessel sends lymph next to a lymph node
- [ ] Lymph node covering: a capsule of dense connective tissue
- [ ] Lymph node sends lymph out through efferent vessels at the hilum, on toward a trunk
- [ ] Node sketch: capsule outside, cortex under it, medulla in the center, hilum on the indented side
- [ ] Afferent vessels enter through the capsule and efferent vessels leave at the hilum
- [ ] Lymph node keeps flow one way because lymph comes in only through the afferent vessels and leaves only through the efferent vessels

Check: Afferent vessels carry lymph into the node and efferent vessels carry it out. Check that your arrows match.

107 Primary and secondary organs

PROMPT A · SORTING BOX

- [ ] Primary column: red bone marrow and thymus
- [ ] Secondary column: lymph nodes, spleen, tonsils and MALT
- [ ] Red bone marrow: spongy bone of the axial skeleton and limb girdles; makes all lymphocytes and matures B cells
- [ ] Thymus: mediastinum, behind the sternum; matures T cells
- [ ] Lymph nodes: along the collecting vessels; filter lymph
- [ ] Spleen: upper left abdomen, behind the stomach; filters blood
- [ ] Tonsils: a ring around the entrance to the throat; guard against inhaled and swallowed germs
- [ ] MALT: mucous membranes lining the body's tracts; guards those surfaces
- [ ] Primary heading: where lymphocytes form and mature
- [ ] Secondary heading: where mature lymphocytes meet invaders

Check: The spleen is the largest lymphatic organ, but it is secondary. Only the red bone marrow and the thymus are primary.

PROMPT B · BODY MAP

- [ ] Red bone marrow placed in the spongy bone of the axial skeleton and limb girdles, such as the sternum, vertebrae and hip bones
- [ ] Thymus in the mediastinum, behind the sternum and above the heart
- [ ] Lymph nodes clustered in the neck, armpits and groin
- [ ] Spleen in the upper left abdomen, behind the stomach and under the lower left ribs
- [ ] Tonsils around the back of the mouth and throat
- [ ] MALT along the linings of the digestive, respiratory, urinary and reproductive tracts
- [ ] Red bone marrow and thymus share one color, the four secondary organs share another
- [ ] Roles: marrow makes lymphocytes and matures B cells, thymus matures T cells, nodes filter lymph, spleen filters blood, tonsils and MALT guard the linings
- [ ] Arrows run from the marrow and thymus out to the nodes, spleen, tonsils and MALT, carrying mature lymphocytes

Check: Only the red bone marrow and the thymus get the primary color. Check that the spleen is in the secondary color.

108 Lymphatic disorders

PROMPT A · DISORDER MAP

- [ ] Edema in the tissue space, swollen with fluid
- [ ] Lymphedema on the leg vessel, which is blocked or missing, with the leg swollen
- [ ] Lymphadenopathy on the neck nodes, drawn enlarged
- [ ] Lymphoma in lymphatic tissue such as the nodes, as cancer of the lymphocytes
- [ ] Splenomegaly on the spleen, drawn enlarged
- [ ] Tonsillitis on the tonsils, infected and inflamed
- [ ] Circles on edema and lymphedema
- [ ] Edema: fluid leaks out faster than healthy lymphatics can drain it. Lymphedema: the vessels themselves are blocked or missing, so the swelling lasts

Check: Edema is a problem in the tissue space, while lymphedema is a problem in the lymphatic vessels. Check that your two labels sit on different structures.

PROMPT B · DISORDER TABLE

- [ ] Edema: interstitial space; fluid leaks out faster than the lymphatics can drain it
- [ ] Lymphedema: lymphatic vessels; long-term swelling from blocked or missing vessels
- [ ] Lymphadenopathy: lymph nodes; any enlargement of the nodes
- [ ] Lymphoma: lymphatic tissue; cancer of the lymphocytes
- [ ] Splenomegaly: spleen; the spleen is enlarged
- [ ] Tonsillitis: tonsils; infected and inflamed
- [ ] Star on splenomegaly
- [ ] Organ at risk: the spleen, which can rupture from a blow

Check: Lymphadenopathy is any enlarged node, while lymphoma is a cancer of lymphocytes. Check that your two rows say different things.

109 Four-layer gut wall

PROMPT A · LABELED CROSS SECTION

- [ ] Rings in order from the lumen out: mucosa, submucosa, muscularis, serosa
- [ ] Mucosa split into epithelium next to the lumen, then lamina propria, then a thin muscularis mucosae
- [ ] Mucosa main tissue: epithelium, with areolar connective tissue under it
- [ ] Submucosa main tissue: areolar connective tissue (OpenStax calls it dense connective tissue)
- [ ] Muscularis main tissue: smooth muscle, inner circular layer and outer longitudinal layer
- [ ] Serosa main tissue: a serous membrane, simple squamous epithelium (mesothelium) over a thin layer of loose connective tissue
- [ ] Serosa shaded as the visceral peritoneum
- [ ] On the esophagus above the diaphragm, a fibrous adventitia replaces the serosa

Check: The circular layer is the inner one and the longitudinal layer is the outer one. Check that you did not swap them.

PROMPT B · COMPARE TABLE

- [ ] Rows in order from the lumen: mucosa, submucosa, muscularis, serosa
- [ ] Mucosa: epithelium; holds the lamina propria and the thin muscularis mucosae
- [ ] Submucosa: areolar connective tissue (OpenStax calls it dense connective tissue); holds blood and lymphatic vessels and the submucosal plexus
- [ ] Muscularis: smooth muscle; holds an inner circular layer, an outer longitudinal layer and the myenteric plexus between them
- [ ] Serosa: serous membrane, also called the visceral peritoneum
- [ ] Esophagus: above the diaphragm, a fibrous adventitia takes the place of the serosa
- [ ] Stomach: an extra inner oblique layer gives it three muscle layers instead of two

Check: The muscularis mucosae is a thin layer inside the mucosa, not the main muscularis. Keep them in separate rows.

110 Enteric nerve plexuses

PROMPT A · LABELED WALL WEDGE

- [ ] Lumen at the bottom with the mucosa lining it
- [ ] Layers in order from the lumen out: mucosa, submucosa, inner circular muscle, outer longitudinal muscle, serosa
- [ ] Submucosal plexus drawn as a dotted network inside the submucosa
- [ ] Myenteric plexus drawn as a dotted network between the circular and longitudinal muscle
- [ ] Submucosal plexus controls secretion
- [ ] Myenteric plexus controls movement
- [ ] The two together make up the enteric nervous system

Check: The myenteric plexus sits between the two muscle layers, not between the submucosa and the muscle. Check where your dots are.

PROMPT B · PLEXUS COMPARE TABLE

- [ ] Submucosal plexus sits in the submucosa
- [ ] Submucosal plexus has the mucosa just inside it and the inner circular muscle just outside it
- [ ] Submucosal plexus controls the secretions of the gut
- [ ] Myenteric plexus sits in the muscularis, between the two muscle layers
- [ ] Myenteric plexus has the inner circular muscle just inside it and the outer longitudinal muscle just outside it
- [ ] Myenteric plexus controls the movement of the gut
- [ ] Title above the table: enteric nervous system
- [ ] Circle on the myenteric plexus

Check: Secretion goes with the submucosal plexus and movement goes with the myenteric plexus. Check that you did not swap them.

111 Peritoneum and folds

PROMPT A · LABELED SAGITTAL VIEW

- [ ] Parietal peritoneum lining the diaphragm, the anterior wall and the posterior wall
- [ ] Visceral peritoneum covering the surface of the organs
- [ ] Peritoneal cavity as the thin space between the two layers
- [ ] Greater omentum hangs like an apron from the greater curvature of the stomach over the intestines and attaches to the transverse colon
- [ ] Lesser omentum connects the stomach and duodenum to the liver
- [ ] Falciform ligament attaches the liver to the anterior abdominal wall and the diaphragm
- [ ] Mesentery binds the jejunum and ileum to the posterior abdominal wall
- [ ] Mesocolon binds the transverse colon to the posterior abdominal wall

Check: The falciform ligament reaches the front wall, while the mesentery and mesocolon reach the back wall. Check which wall each fold touches.

PROMPT B · CONNECTION MAP

- [ ] Greater omentum: line from the stomach to the transverse colon, hanging down over the intestines
- [ ] Lesser omentum: lines from the stomach and the duodenum to the liver
- [ ] Falciform ligament: lines from the liver to the anterior abdominal wall and the diaphragm
- [ ] Mesentery: line from the jejunum and ileum to the posterior abdominal wall
- [ ] Mesocolon: lines from the transverse colon and the sigmoid colon to the posterior abdominal wall
- [ ] Parietal peritoneum as the outer curve, lining the body wall
- [ ] Visceral peritoneum as the inner curve, covering the organ
- [ ] Peritoneal cavity as the thin, fluid-filled space between them

Check: The mesocolon holds the transverse and sigmoid colon, not the ascending or descending colon. Check your mesocolon lines.

112 Retroperitoneal organs

PROMPT A · LABELED CROSS SECTION

- [ ] Parietal peritoneum drawn as a line along the posterior wall, near the back of your box
- [ ] Kidneys behind the parietal peritoneum, against the back wall
- [ ] Pancreas behind the parietal peritoneum
- [ ] Duodenum behind the parietal peritoneum
- [ ] Ascending or descending colon behind the parietal peritoneum
- [ ] Jejunum hanging forward into the peritoneal cavity on the mesentery
- [ ] Jejunum wrapped all the way around in visceral peritoneum
- [ ] Kidneys, pancreas, duodenum and colon all in one color
- [ ] Jejunum is not retroperitoneal because it hangs inside the peritoneal cavity on a fold instead of sitting behind the parietal peritoneum

Check: Retroperitoneal organs sit behind the parietal peritoneum, against the back wall. If your kidney is inside the peritoneal cavity, move it.

PROMPT B · SORTING BOX

- [ ] Behind the peritoneum: kidneys, pancreas, duodenum, ascending colon and descending colon
- [ ] Inside the cavity: stomach, liver, jejunum, ileum, transverse colon and sigmoid colon
- [ ] Stomach: greater omentum and lesser omentum
- [ ] Liver: falciform ligament and lesser omentum
- [ ] Jejunum and ileum: mesentery
- [ ] Transverse colon and sigmoid colon: mesocolon
- [ ] Retroperitoneal means the organ sits behind the parietal peritoneum, against the back wall, outside the peritoneal cavity

Check: The transverse and sigmoid colon hang on a mesocolon, so only the ascending and descending colon are retroperitoneal.

113 Mouth, pharynx, esophagus

PROMPT A · LABELED SAGITTAL VIEW

- [ ] Hard palate as the bony front of the roof of the mouth, soft palate as the muscular back part
- [ ] Uvula hanging from the back edge of the soft palate
- [ ] Fauces as the opening from the mouth into the oropharynx
- [ ] Palatine tonsil in the side wall of the oropharynx between the palatoglossal and palatopharyngeal arches, lingual tonsil at the base of the tongue
- [ ] Nasopharynx above the soft palate, oropharynx behind the mouth, laryngopharynx from the epiglottis down to the esophagus
- [ ] Esophagus runs from the laryngopharynx down to the diaphragm line
- [ ] Esophageal hiatus is the opening in the diaphragm where the esophagus passes through
- [ ] Boundaries of the oral cavity: the lips in front, the cheeks on the sides, the hard and soft palate as the roof, the mylohyoid muscle as the floor with the tongue resting on it, and the fauces at the back, opening into the oropharynx
- [ ] Soft palate and uvula lifted against the back of the throat, closing off the nasopharynx

Check: Food passes through the oropharynx and laryngopharynx only. The nasopharynx is closed off by the uvula during a swallow.

PROMPT B · FRONT VIEW AND CHAIN

- [ ] Lips at the front and cheeks on the sides
- [ ] Hard palate at the front of the roof and soft palate behind it
- [ ] Uvula hanging in the midline from the soft palate
- [ ] Fauces as the opening at the back of the mouth into the oropharynx
- [ ] Palatine tonsils on each side, between the palatal arches
- [ ] Lingual tonsil at the base of the tongue
- [ ] Food chain in order: fauces, oropharynx, laryngopharynx, esophagus, esophageal hiatus, stomach
- [ ] Nasopharynx sits in a side box off the oropharynx, outside the food path
- [ ] Hard palate bones: the maxillae and the palatine bones
- [ ] Esophagus length: about 25 cm (10 inches)

Check: The nasopharynx is not on the food path. It belongs in a side box, not in the chain.

114 Major canal sphincters

PROMPT A · LABELED CANAL OUTLINE

- [ ] Upper esophageal sphincter between the laryngopharynx and the esophagus
- [ ] Lower esophageal (cardiac) sphincter between the esophagus and the stomach
- [ ] Pyloric sphincter between the stomach and the duodenum
- [ ] Ileocecal sphincter between the ileum and the cecum of the large intestine
- [ ] Internal and external anal sphincters ringing the anal canal, with the outside of the body past the anus
- [ ] Pyloric sphincter: smooth muscle
- [ ] Ileocecal sphincter: smooth muscle
- [ ] Internal anal sphincter: smooth muscle, involuntary
- [ ] External anal sphincter: skeletal muscle, voluntary, and colored

Check: The internal anal sphincter is smooth muscle you do not control. Only the external anal sphincter gets colored.

PROMPT B · SPHINCTER TABLE

- [ ] Rows in order: upper esophageal, lower esophageal, pyloric, ileocecal, internal anal, external anal
- [ ] Upper esophageal: laryngopharynx before, esophagus after
- [ ] Lower esophageal: esophagus before, stomach after, also called the cardiac sphincter
- [ ] Pyloric: stomach before, duodenum after
- [ ] Ileocecal: ileum before, cecum after, also called the ileocecal valve
- [ ] Internal and external anal: anal canal before, the outside of the body after the anus
- [ ] Stars on the internal and external anal sphincters
- [ ] The external anal sphincter is the one you control by choice

Check: The pyloric sphincter opens into the duodenum, not the jejunum. Check the region after it.

115 Stomach regions and features

PROMPT A · LABELED ANTERIOR VIEW

- [ ] Esophagus entering at the cardia and duodenum leaving past the pyloric sphincter
- [ ] Cardia around the opening from the esophagus
- [ ] Fundus bulging up above and to the left of the cardia
- [ ] Body as the large central part
- [ ] Pyloric part in order: pyloric antrum, pyloric canal, pylorus
- [ ] Pyloric sphincter as a ring between the pylorus and the duodenum
- [ ] Lesser curvature along the short, concave right (medial) border facing the liver; greater curvature along the long, convex left (lateral) border
- [ ] Rugae drawn as folds of the inner lining in the window
- [ ] Muscle layers from inside to outside: oblique, circular, longitudinal
- [ ] The oblique layer is the one the rest of the tube does not have

Check: The fundus bulges up beside the esophagus, not at the bottom of the stomach. Check where yours sits.

PROMPT B · REGION TABLE

- [ ] Cardia: around the opening from the esophagus
- [ ] Fundus: rounded bulge above and to the left of the cardia
- [ ] Body: the large central portion
- [ ] Pyloric part: links the body to the duodenum, made of the pyloric antrum, pyloric canal and pylorus
- [ ] Lesser curvature: the short, concave right (medial) border, facing the liver
- [ ] Greater curvature: the long, convex left (lateral) border
- [ ] Rugae: folds of the inner lining that flatten as the stomach fills
- [ ] Wall strip from the lumen out: inner oblique, middle circular, outer longitudinal

Check: The lesser curvature is the short, inward curving border on the right, facing the liver. The greater curvature is the long, outward bulging border on the left.

116 Small intestine and surface area

PROMPT A · LABELED MODEL VIEW

- ☐ Duodenum about 25 cm (10 inches), curving from the pyloric sphincter
- ☐ Jejunum about 1 m (3 feet)
- ☐ Ileum about 2 m (6 feet), ending at the ileocecal sphincter
- ☐ Mesentery fanning from the jejunum and ileum to the posterior wall
- ☐ Circular fold drawn as a permanent ridge of the lining
- ☐ Villus with a lacteal running up its center
- ☐ Microvilli on the surface of one absorptive cell, forming the brush border
- ☐ Duodenum shaded as the mostly retroperitoneal region
- ☐ Surface features numbered 1 circular folds, 2 villi, 3 microvilli

Check: The ileum is the longest part, not the jejunum. Check your lengths: about 25 cm (10 inches), 1 m (3 feet) and 2 m (6 feet).

PROMPT B · ZOOM SEQUENCE

- ☐ Duodenum about 25 cm (10 inches), jejunum about 1 m (3 feet), ileum about 2 m (6 feet)
- ☐ First zoom: a circular fold made of mucosa and submucosa
- ☐ Second zoom: one finger-shaped villus with a lacteal in its center
- ☐ Third zoom: one absorptive cell with microvilli forming the brush border
- ☐ Each zoom drawn smaller than the one before it
- ☐ Lacteal absorbs dietary fats
- ☐ Ileum joins the large intestine at the ileocecal sphincter

Check: Circular folds are the biggest and microvilli the smallest. Microvilli sit on one cell, so they belong in the last zoom.

117 Large intestine features

PROMPT A · LABELED ANTERIOR VIEW

- ☐ Ileum entering at the ileocecal valve
- ☐ Cecum as the pouch just below the valve, with the vermiform appendix hanging off it
- ☐ Ascending colon up the right side, transverse colon across the top, descending colon down the left side
- ☐ Sigmoid colon curving into the pelvis, then the rectum and the anal canal
- ☐ Internal and external anal sphincters around the anal canal
- ☐ Teniae coli as bands of muscle running along the transverse colon
- ☐ Haustra as the pouches between the bands
- ☐ Omental appendices as small fat pouches hanging from the teniae coli
- ☐ Where they sit: the rectum is the straight part in the pelvis, in front of the sacrum and coccyx; the anal canal is the last part, opening at the anus
- ☐ Transverse colon and sigmoid colon colored as the parts on a mesocolon

Check: The transverse and sigmoid colon hang on a mesocolon. The ascending and descending colon should not be colored.

PROMPT B · REGION TABLE

- ☐ Rows in order: cecum, ascending colon, transverse colon, descending colon, sigmoid colon, rectum, anal canal
- ☐ Cecum: ileum before, ascending colon after; it carries the appendix
- ☐ Ascending colon: cecum before, transverse colon after; runs up the right side
- ☐ Transverse colon: ascending colon before, descending colon after; crosses the abdomen
- ☐ Descending colon: transverse colon before, sigmoid colon after; runs down the left side
- ☐ Sigmoid colon: descending colon before, rectum after; S-shaped curve into the pelvis
- ☐ Rectum: sigmoid colon before, anal canal after; straight, in front of the sacrum and coccyx
- ☐ Anal canal: rectum before, anus after; ringed by the anal sphincters
- ☐ Cecum sketch with the appendix attached to it
- ☐ Colon piece with three teniae coli, haustra between them and omental appendices on the bands

Check: There are three teniae coli, not two or four. Count the bands on your colon.

118 Trace the pathway of food

PROMPT A · FLOW CHAIN

- ☐ Starts at the mouth and ends at the anus
- ☐ Pharynx shown as the oropharynx then the laryngopharynx, with no nasopharynx
- ☐ Esophagus then stomach
- ☐ Small intestine as duodenum, jejunum, ileum
- ☐ Large intestine as cecum, ascending, transverse, descending and sigmoid colon, rectum, anal canal
- ☐ Rings at the upper esophageal, lower esophageal, pyloric and ileocecal sphincters, and the anal sphincters
- ☐ Words under the boxes: chewed in the mouth, swallowed in the pharynx, carried down the esophagus, churned in the stomach
- ☐ More words: bile and pancreatic juice added in the duodenum, nutrients absorbed in the jejunum and ileum, water absorbed and feces formed in the colon, feces stored in the rectum, feces eliminated at the anus

Check: The nasopharynx is not on the food path, and the cecum comes right after the ileum. Check those two spots in your chain.

PROMPT B · BODY OUTLINE MAP

- ☐ Diaphragm drawn with the esophagus passing through it and the stomach just below it
- ☐ Mouth numbered first, then oropharynx, laryngopharynx and esophagus
- ☐ Stomach in the upper left abdomen, under the diaphragm
- ☐ Duodenum, jejunum and ileum numbered in that order, coiled in the middle
- ☐ Cecum in the lower right, then ascending, transverse, descending and sigmoid colon framing the abdomen
- ☐ Rectum, anal canal and anus numbered last
- ☐ Circle around the small intestine, where most absorption happens
- ☐ Box around the large intestine, where water is absorbed and feces form

Check: Most absorption happens in the small intestine. The large intestine gets the box, not the circle.

119 Clinical anatomy of canal disorders

PROMPT A · DISORDER MAP

- [ ] GERD at the lower esophageal sphincter, with stomach contents flowing back up into the esophagus
- [ ] Hiatal hernia at the esophageal hiatus, with part of the stomach pushed up above the diaphragm
- [ ] Peptic ulcer as a sore in the wall of the stomach or duodenum
- [ ] Appendicitis on the vermiform appendix, drawn swollen
- [ ] Diverticulosis on the colon, with small pouches bulging outward
- [ ] Inflammatory bowel disease on the intestinal wall, drawn inflamed
- [ ] Hemorrhoids as swollen veins in the wall of the anal canal and rectum
- [ ] Colorectal cancer as a tumor in the colon or rectum
- [ ] Lower esophageal sphincter: heartburn felt behind the breastbone
- [ ] Appendix: pain felt in the lower right abdomen

Check: A hiatal hernia is the stomach pushing up through the diaphragm, while GERD is backflow through a weak sphincter. Keep them on separate spots.

PROMPT B · COMPARE TABLE

- [ ] GERD: lower esophageal sphincter and esophagus; a weak sphincter lets stomach contents flow back and irritate the esophagus
- [ ] Hiatal hernia: esophageal hiatus of the diaphragm; part of the stomach pushes up through it
- [ ] Peptic ulcer: wall of the stomach or duodenum; an eroded sore
- [ ] Appendicitis: vermiform appendix; inflamed, and rupture can spread infection to the peritoneum
- [ ] Diverticulosis: colon wall; small pouches bulge out through weak spots
- [ ] Inflammatory bowel disease: intestinal wall; chronic inflammation, as in Crohn disease and ulcerative colitis
- [ ] Hemorrhoids: anal canal and rectum; swollen, dilated veins
- [ ] Colorectal cancer: colon or rectum; a malignant tumor
- [ ] Wall drawn from inside out: mucosa, submucosa, muscularis, serosa
- [ ] Ulcer starts in the mucosa, erodes through the submucosa and muscularis, then breaks through the serosa into the peritoneal cavity

Check: A peptic ulcer starts on the inside, in the mucosa, and works outward. Check that your drawing does not start at the serosa.

120 Teeth and tongue

PROMPT A · LABELED MOUTH VIEW

- [ ] Upper and lower teeth drawn in the open mouth
- [ ] Lingual frenulum as the midline fold under the lifted tongue, anchoring it to the floor of the mouth
- [ ] Papillae on the upper surface of the tongue in the side view
- [ ] Intrinsic muscles drawn inside the body of the tongue
- [ ] Extrinsic muscles starting outside the tongue and running into it
- [ ] Lingual glands inside the tongue
- [ ] Extrinsic muscle arrows show the tongue moving out and back, up and down, and side to side
- [ ] Intrinsic muscles change the shape and size of the tongue for speech and swallowing

Check: The extrinsic muscles move the whole tongue, while the intrinsic muscles change its shape. Check that your movement arrows belong to the extrinsic group.

PROMPT B · STRUCTURE TABLE

- [ ] Teeth: in the upper and lower jaws; cut and grind food
- [ ] Tongue: in the floor of the mouth; skeletal muscle covered with mucous membrane that moves food for chewing and swallowing
- [ ] Lingual frenulum: midline under the tongue; anchors the tongue to the floor of the mouth
- [ ] Papillae: on the upper surface of the tongue; most kinds carry taste buds, and the filiform papillae have none and help grip food
- [ ] Intrinsic tongue muscles: within the tongue; change its shape and size for speech and swallowing
- [ ] Extrinsic tongue muscles: start outside the tongue; move it out and back, up and down, and side to side
- [ ] Lingual glands: in the tongue; secrete mucus and watery fluid
- [ ] Circles on the two muscle groups, with extrinsic marked as starting outside and intrinsic as sitting within

Check: Extrinsic means it starts outside the tongue, and intrinsic means it sits within it. Check that you did not swap them.

121 Salivary glands

PROMPT A · LABELED FACE VIEW

- [ ] Parotid gland in front of and below the ear
- [ ] Parotid duct crossing the cheek and opening inside the mouth by the second upper molar
- [ ] Submandibular gland along the inner side of the mandible
- [ ] Submandibular duct opening under the tongue, near the lingual frenulum
- [ ] Sublingual gland beneath the tongue, above the submandibular gland
- [ ] Sublingual ducts opening onto the floor of the mouth
- [ ] Each gland and its own opening drawn in the same color
- [ ] Parotid gland lies over the masseter muscle

Check: The parotid duct opens in the cheek beside the second upper molar, not under the tongue. Check where yours ends.

PROMPT B · LOCATION TABLE

- [ ] Parotid: in front of and below the ear; next to the masseter muscle; opens by the second upper molar
- [ ] Submandibular: floor of the mouth along the inner side of the mandible; next to the mandible; opens near the lingual frenulum
- [ ] Sublingual: beneath the tongue; above the submandibular gland; opens onto the floor of the mouth
- [ ] Floor of the mouth sketch with the tongue lifted and the lingual frenulum in the midline
- [ ] Submandibular duct openings marked beside the lingual frenulum
- [ ] Sublingual duct openings marked along the floor of the mouth

Check: Two of the three ducts open under the tongue. Only the parotid duct opens in the cheek.

122 Liver gross and lobule anatomy

PROMPT A · LABELED VIEWS AND ZOOM

- [ ] Front view: large right lobe and smaller left lobe
- [ ] Falciform ligament between the right and left lobes
- [ ] Ligamentum teres in the free lower edge of the falciform ligament
- [ ] Quadrate lobe on the inferior surface between the gallbladder and the ligamentum teres; caudate lobe on the posterior surface beside the inferior vena cava
- [ ] Hepatocytes drawn in plates
- [ ] Sinusoids running between the plates
- [ ] Bile canaliculi as tiny channels between the hepatocytes
- [ ] Portal triad at a corner of the lobule: bile duct, branch of the hepatic artery, branch of the hepatic portal vein
- [ ] Ligamentum teres is a remnant of the fetal umbilical vein

Check: The ligamentum teres comes from the umbilical vein, not an umbilical artery. Check your fetal vessel.

PROMPT B · GROSS TO MICRO TABLE

- [ ] Right lobe: by eye; the larger principal lobe, separated from the left lobe by the falciform ligament
- [ ] Left lobe: by eye; the smaller principal lobe
- [ ] Quadrate and caudate lobes: by eye; quadrate on the inferior surface between the gallbladder and the ligamentum teres, caudate on the posterior surface beside the inferior vena cava
- [ ] Falciform ligament: by eye; fold of peritoneum that hangs the liver from the anterior wall and diaphragm
- [ ] Ligamentum teres: by eye; fibrous cord in the free edge of the falciform ligament, left from the fetal umbilical vein
- [ ] Hepatocytes: slide; arranged in plates; the working liver cells that make bile
- [ ] Sinusoids: slide; between the plates; wide, leaky capillaries
- [ ] Bile canaliculi: slide; between the hepatocytes; collect the bile
- [ ] Portal triad: slide; at the corner of each lobule; a bile duct, a hepatic artery branch and a hepatic portal vein branch
- [ ] Triad sketch: arrows into the liver on the artery and portal vein branches, arrow out on the bile duct

Check: In the portal triad two vessels bring blood in and only the bile duct carries bile out. Check your arrows.

123 Gallbladder and bile pathway

PROMPT A · LABELED BILIARY TREE

- [ ] Gallbladder on the underside of the liver: fundus at the broad end, body in the middle, neck at the tapered end
- [ ] Bile canaliculi as tiny channels between hepatocytes
- [ ] Right and left hepatic ducts merging into the common hepatic duct
- [ ] Cystic duct running from the neck of the gallbladder to join the common hepatic duct
- [ ] Common bile duct formed where the common hepatic and cystic ducts join
- [ ] Pancreatic duct joining the common bile duct at the hepatopancreatic ampulla
- [ ] Sphincter of Oddi around the ampulla
- [ ] Major duodenal papilla where the ampulla opens into the duodenum
- [ ] Arrows run canaliculi, hepatic ducts, common hepatic duct, common bile duct, ampulla, duodenum
- [ ] Side loop of arrows into and back out of the gallbladder through the cystic duct

Check: The cystic duct joins the common hepatic duct to make the common bile duct. Check that your cystic duct does not run straight to the duodenum.

PROMPT B · FLOW CHAIN

- [ ] Chain starts at the hepatocyte, then the bile canaliculi
- [ ] Next: right and left hepatic ducts, then the common hepatic duct
- [ ] Next: common bile duct, then the hepatopancreatic ampulla
- [ ] Last: major duodenal papilla into the duodenum
- [ ] Side arrow through the cystic duct to the gallbladder and back
- [ ] Gallbladder sketch with the fundus at the broad end, body in the middle and neck at the tapered end
- [ ] Pancreatic duct joins at the hepatopancreatic ampulla
- [ ] Sphincter of Oddi controls flow at the ampulla into the duodenum
- [ ] A box for the bile ducts of the portal triads, between the bile canaliculi and the right and left hepatic ducts

Check: The cystic duct and gallbladder are a side trip, not a step every drop takes. Bile can go straight down the common bile duct.

124 Pancreas structure

PROMPT A · LABELED VIEW AND ZOOM

- [ ] Head of the pancreas sitting in the curve of the duodenum
- [ ] Body behind the stomach and tail tapering off to the left
- [ ] Main pancreatic duct running the length of the gland
- [ ] Accessory duct emptying separately into the duodenum at the minor duodenal papilla, a little above the major papilla
- [ ] Common bile duct joining the main duct at the hepatopancreatic ampulla
- [ ] Sphincter of Oddi around the ampulla
- [ ] Major duodenal papilla where the ampulla opens into the duodenum
- [ ] Main duct is the duct of Wirsung; accessory duct is the duct of Santorini
- [ ] Acini labeled exocrine, about 99 percent of the gland
- [ ] Islet labeled endocrine, about 1 percent of the gland

Check: Only the main duct (Wirsung) joins the common bile duct. The accessory duct (Santorini) empties on its own.

PROMPT B · COMPARE TABLE

- [ ] Head: in the curve of the duodenum
- [ ] Body: behind the stomach. Tail: the tapered end
- [ ] Main pancreatic duct: runs through the gland; duct of Wirsung; carries pancreatic juice and joins the common bile duct
- [ ] Accessory duct: drains part of the head; duct of Santorini; empties separately into the duodenum at the minor duodenal papilla
- [ ] Major duodenal papilla: raised opening in the duodenal lining; where bile and pancreatic juice enter
- [ ] Sphincter of Oddi: ring at the hepatopancreatic ampulla; also called the hepatopancreatic sphincter; controls flow into the duodenum
- [ ] Acini: exocrine clusters; make pancreatic juice, about 99 percent of the gland
- [ ] Pancreatic islets: scattered among the acini; also called the islets of Langerhans; endocrine clusters that make hormones, about 1 percent of the gland
- [ ] Sketch shows two ducts entering the duodenum, with the main duct circled as the one joining the common bile duct

Check: Only the main duct (Wirsung) joins the common bile duct. The accessory duct (Santorini) empties on its own.

125 Clinical anatomy of accessory disorders

PROMPT A · DISORDER MAP

- [ ] Gallstones in the gallbladder and bile ducts
- [ ] Cholecystitis on the gallbladder
- [ ] Hepatitis and cirrhosis on the liver
- [ ] Jaundice on the liver and bile ducts, where a bilirubin buildup starts
- [ ] Pancreatitis on the pancreas
- [ ] Mumps on the parotid gland in the face view
- [ ] Gallstone stuck at the hepatopancreatic ampulla
- [ ] Arrows show bile backed up in the common bile duct and pancreatic juice backed up in the pancreatic duct
- [ ] This blockage can trigger pancreatitis

Check: Mumps swells the parotid gland, not the pancreas. Check where your mumps label sits.

PROMPT B · COMPARE TABLE

- [ ] Gallstones: gallbladder; hard deposits that can block the bile ducts
- [ ] Cholecystitis: gallbladder; inflamed, often from a blocking stone
- [ ] Hepatitis: liver; inflamed, often from a virus
- [ ] Cirrhosis: liver; healthy tissue replaced by scar tissue
- [ ] Jaundice: liver or bile ducts; bilirubin builds up and yellows the skin and eyes
- [ ] Pancreatitis: pancreas; inflamed
- [ ] Mumps: parotid glands; a virus inflames and enlarges them
- [ ] Sketch shows the bile duct and pancreatic duct meeting at the ampulla with a stone in it
- [ ] The two flows that stop are bile and pancreatic juice

Check: Cholecystitis is the gallbladder, while hepatitis and cirrhosis are the liver. Check each organ name.

126 Abdominal aorta and portal system

PROMPT A · LABELED ANTERIOR VIEW

- [ ] Abdominal aorta with the inferior vena cava beside it on the body's right
- [ ] Celiac trunk leaving the front of the aorta just below the diaphragm, splitting into the left gastric, splenic and common hepatic arteries
- [ ] Superior mesenteric artery just below the celiac trunk
- [ ] Paired renal arteries to the kidneys, then paired gonadal arteries below them
- [ ] Inferior mesenteric artery below the gonadals
- [ ] Aorta splitting at the bottom into the right and left common iliac arteries
- [ ] Splenic vein and superior mesenteric vein joining to form the hepatic portal vein, which enters the liver
- [ ] Inferior mesenteric vein draining into the splenic vein (its usual pattern)
- [ ] Hepatic veins leaving the liver and draining into the inferior vena cava
- [ ] Numbers 1 celiac trunk, 2 superior mesenteric, 3 inferior mesenteric. Portal system: blood passes through two capillary beds in a row, in the intestine and then in the liver

Check: The hepatic portal vein carries blood into the liver and the hepatic veins carry it out. Check that you did not swap them.

PROMPT B · BRANCH AND DRAINAGE TABLE

- [ ] Celiac trunk: unpaired; branches off the abdominal aorta
- [ ] Left gastric, splenic and common hepatic arteries: each a single artery; branch off the celiac trunk
- [ ] Superior mesenteric and inferior mesenteric arteries: unpaired; branch off the aorta
- [ ] Renal and gonadal arteries: paired; branch off the aorta
- [ ] Hepatic portal vein: unpaired; drains into the liver
- [ ] Splenic and superior mesenteric veins: unpaired; join to form the hepatic portal vein
- [ ] Inferior mesenteric vein: unpaired; usually drains into the splenic vein
- [ ] Hepatic veins: drain the liver into the inferior vena cava
- [ ] Glucose chain: intestinal capillary, superior mesenteric vein, hepatic portal vein, liver sinusoids, central vein, hepatic veins, inferior vena cava, right atrium
- [ ] Circles on the intestinal capillaries and the liver sinusoids

Check: The three unpaired branches are the celiac trunk, superior mesenteric and inferior mesenteric. The renal and gonadal arteries are paired.

127 Lower limb vessels and nerves

PROMPT A · FRONT AND BACK VIEWS

- [ ] Femoral artery running down the front of the thigh with the femoral vein beside it on the inner side
- [ ] Femoral nerve on the outer side of the femoral artery at the top of the thigh
- [ ] Great saphenous vein running up the inner side of the leg and thigh to the groin
- [ ] Sciatic nerve running down the back of the thigh
- [ ] Sciatic nerve splitting above the knee into the tibial nerve and the common fibular nerve
- [ ] Common fibular nerve heading to the outer side, tibial nerve continuing down the back of the leg
- [ ] Posterior tibial artery in the back of the leg
- [ ] Femoral vein deep, great saphenous vein superficial
- [ ] Femoral nerve from the lumbar plexus; sciatic, tibial and common fibular nerves from the sacral plexus

Check: The great saphenous vein is superficial and runs on the inner side of the limb. The femoral vein is the deep one.

PROMPT B · STRUCTURE-TO-REGION TABLE

- [ ] Femoral artery: artery; thigh
- [ ] Femoral vein: vein; thigh; deep
- [ ] Great saphenous vein: vein; both thigh and leg; superficial
- [ ] Posterior tibial artery: artery; leg
- [ ] Femoral nerve: nerve; thigh
- [ ] Sciatic nerve: nerve; thigh
- [ ] Tibial and common fibular nerves: nerves; leg, after the sciatic splits just above the knee
- [ ] Lumbar plexus box branches to the femoral nerve
- [ ] Sacral plexus box branches to the sciatic nerve, which branches into the tibial and common fibular nerves

Check: The tibial and common fibular nerves come off the sciatic nerve, not straight off the plexus. Check your branch lines.

128 Major endocrine glands and locations

PROMPT A · LABELED BODY MAP

- [ ] Hypothalamus in the floor and walls of the third ventricle
- [ ] Pituitary gland in the sella turcica of the sphenoid, hanging below the hypothalamus
- [ ] Pineal gland at the back of the roof of the third ventricle
- [ ] Thyroid gland in the front of the neck, just below the larynx, with its lobes beside the trachea and the isthmus crossing in front of it
- [ ] Parathyroid glands on the back of the thyroid
- [ ] Thymus in the mediastinum, behind the sternum
- [ ] Adrenal glands capping the top of each kidney
- [ ] Pancreatic islets in the pancreas, behind the stomach
- [ ] Ovaries one on each side of the uterus, testes in the scrotum
- [ ] Star on the thyroid gland, and the thymus noted as large in childhood and shrinking after puberty

Check: The pineal gland sits at the back of the third ventricle, while the pituitary hangs below the hypothalamus. Check that they are not in the same spot.

PROMPT B · LOCATION TABLE

- [ ] Hypothalamus: head; floor and walls of the third ventricle
- [ ] Pituitary gland: head; sella turcica of the sphenoid bone
- [ ] Pineal gland: head; back of the roof of the third ventricle
- [ ] Thyroid gland: neck; lobes beside the trachea and isthmus in front of it, below the larynx
- [ ] Parathyroid glands: neck; back surface of the thyroid
- [ ] Thymus: chest; mediastinum, behind the sternum
- [ ] Adrenal glands: abdomen; top of each kidney
- [ ] Pancreatic islets: abdomen; inside the pancreas, behind the stomach
- [ ] Ovaries: pelvis, beside the uterus. Testes: scrotum
- [ ] Circles on the parathyroids, adrenals, ovaries and testes, with usually four parathyroid glands. Circling the pancreatic islets is also fair, since they are many clusters in one pancreas

Check: There are usually four parathyroid glands, not two. Check your count.

129 Pituitary gross anatomy and lobes

PROMPT A · LABELED SAGITTAL VIEW

- [ ] Hypothalamus above, with the infundibulum as the stalk running down from it
- [ ] Pituitary gland sitting in the sella turcica of the sphenoid bone
- [ ] Anterior lobe (adenohypophysis) in front, posterior lobe (neurohypophysis) behind
- [ ] Nerve fibers running from the hypothalamus down the infundibulum and ending in the posterior lobe
- [ ] Hypophyseal portal vessels running from the hypothalamus to the anterior lobe
- [ ] Anterior lobe: glandular epithelial tissue
- [ ] Posterior lobe: nervous tissue
- [ ] Tumor arrow points up out of the sella turcica
- [ ] Tumor presses on the optic chiasm

Check: The nerve fibers go to the posterior lobe and the blood vessels go to the anterior lobe. Check that you did not swap them.

PROMPT B · TWO-LOBE COMPARE TABLE

- [ ] Anterior lobe: other name adenohypophysis
- [ ] Anterior lobe: glandular epithelial tissue
- [ ] Anterior lobe: linked to the hypothalamus by the hypophyseal portal system, a set of blood vessels
- [ ] Posterior lobe: other name neurohypophysis
- [ ] Posterior lobe: nervous tissue, an extension of the hypothalamus
- [ ] Posterior lobe: linked by a tract of nerve fibers running down the infundibulum
- [ ] Sketch shows the gland in the sella turcica, hanging from the infundibulum below the hypothalamus
- [ ] Sentence: nervous tissue links by nerve fibers, glandular tissue links by blood vessels

Check: The posterior lobe is the nervous one and the anterior lobe is the glandular one. Check that your rows match.

130 Thyroid and parathyroid anatomy

PROMPT A · LABELED VIEWS WITH INSETS

- [ ] Thyroid sitting on the trachea just below the larynx
- [ ] Right and left lobes on each side of the trachea
- [ ] Isthmus crossing the front of the trachea, joining the two lobes
- [ ] Back view with four small parathyroid glands, two on the back of each thyroid lobe
- [ ] Thyroid follicles drawn as hollow spheres
- [ ] Colloid filling the center of each follicle
- [ ] Follicular cells forming the wall of each follicle
- [ ] Parafollicular (C) cells between the follicles or at the outer edge of a follicle wall, not touching the colloid
- [ ] Chief cells labeled in the parathyroid circle
- [ ] Risk: the parathyroids are tiny and embedded on the back of the thyroid, so they can be damaged or removed with it

Check: Colloid fills the inside of the follicle. The C cells sit between follicles or at the outer edge of the wall, and they never touch the colloid.

PROMPT B · ZOOM-IN SEQUENCE

- [ ] First box: neck with the thyroid on the trachea below the larynx
- [ ] Second box: right and left lobes joined by the isthmus across the front of the trachea
- [ ] Third box: follicles as hollow spheres, colloid in the center, follicular cells in the wall
- [ ] Third box: parafollicular (C) cells between the follicles or at the outer edge of a follicle wall, not touching the colloid
- [ ] Side box: parathyroid gland with chief cells labeled
- [ ] Neck box and gland box: model or cadaver
- [ ] Follicle box and parathyroid cell box: slide

Check: Follicles, colloid and C cells are too small to see by eye, so that box can only be the slide.

131 Adrenal gland zones

PROMPT A · LABELED CUTAWAY

- [ ] Adrenal gland capping the top (superior pole) of the kidney
- [ ] Capsule as the thin outer covering
- [ ] Adrenal cortex as the outer bulk of the gland, adrenal medulla as the core
- [ ] Zones from the surface in: zona glomerulosa, zona fasciculata, zona reticularis
- [ ] Glomerulosa drawn thin and fasciculata drawn as the thickest zone
- [ ] Reticularis as the inner zone, against the medulla
- [ ] Cortex: glandular tissue. Medulla: modified nervous tissue
- [ ] Fasciculata cells drawn in straight columns, reticularis cells as a branching network

Check: The order from the surface in is glomerulosa, fasciculata, reticularis, and the middle zone is the thickest. Check your order and widths.

PROMPT B · LAYER TABLE

- [ ] Zona glomerulosa: outer; thin; glandular tissue
- [ ] Zona fasciculata: middle; thickest of the three; glandular tissue
- [ ] Zona reticularis: inner, against the medulla; thinner than the fasciculata; glandular tissue
- [ ] Adrenal medulla: core; modified nervous tissue
- [ ] Kidney imaging often shows the adrenal because the gland rests on the top of the kidney

Check: The medulla is nervous tissue, not glandular like the three cortex zones. Check your tissue column.

132 Pancreas, gonads, and organs with endocrine tissue

PROMPT A · LABELED BODY MAP

- [ ] Pancreas behind the stomach
- [ ] Islets as small clusters scattered among many acini, labeled endocrine, with the acini labeled exocrine
- [ ] Ovaries on each side of the uterus, testes in the scrotum
- [ ] Heart: cardiac muscle cells in the walls of the atria
- [ ] Kidneys: endocrine cells in the renal cortex, including the juxtaglomerular cells
- [ ] Skin: keratinocytes of the epidermis, which begin making a hormone in sunlight
- [ ] Stomach and small intestine: enteroendocrine cells scattered in the mucosa
- [ ] Liver: hepatocytes, the main liver cells
- [ ] Adipose tissue: adipocytes (fat cells)
- [ ] Placenta: placental cells, and it is temporary because it exists only during pregnancy

Check: The islets are the endocrine part and the acini are exocrine. Check that your labels did not swap.

PROMPT B · ORGAN SORTING TABLE

- [ ] Pancreas: behind the stomach; pancreatic islets
- [ ] Ovaries: pelvis, one on each side of the uterus; cells of the ovarian follicles and the corpus luteum
- [ ] Testes: scrotum; interstitial (Leydig) cells between the seminiferous tubules
- [ ] Heart: chest; cardiac muscle cells in the walls of the atria
- [ ] Kidneys: abdomen; endocrine cells in the renal cortex, including the juxtaglomerular cells
- [ ] Skin: body surface; keratinocytes of the epidermis
- [ ] Stomach and small intestine: abdomen; enteroendocrine cells scattered in the mucosa
- [ ] Liver: upper right abdomen; hepatocytes
- [ ] Adipose tissue: under the skin and around organs; adipocytes (fat cells). Placenta: in the uterus during pregnancy; placental cells
- [ ] Circles on the pancreas, ovaries and testes, which are mixed because they do an endocrine job alongside another role

Check: Your notes call only the pancreas, ovaries and testes mixed organs. The heart and kidneys contain endocrine tissue but are not circled.